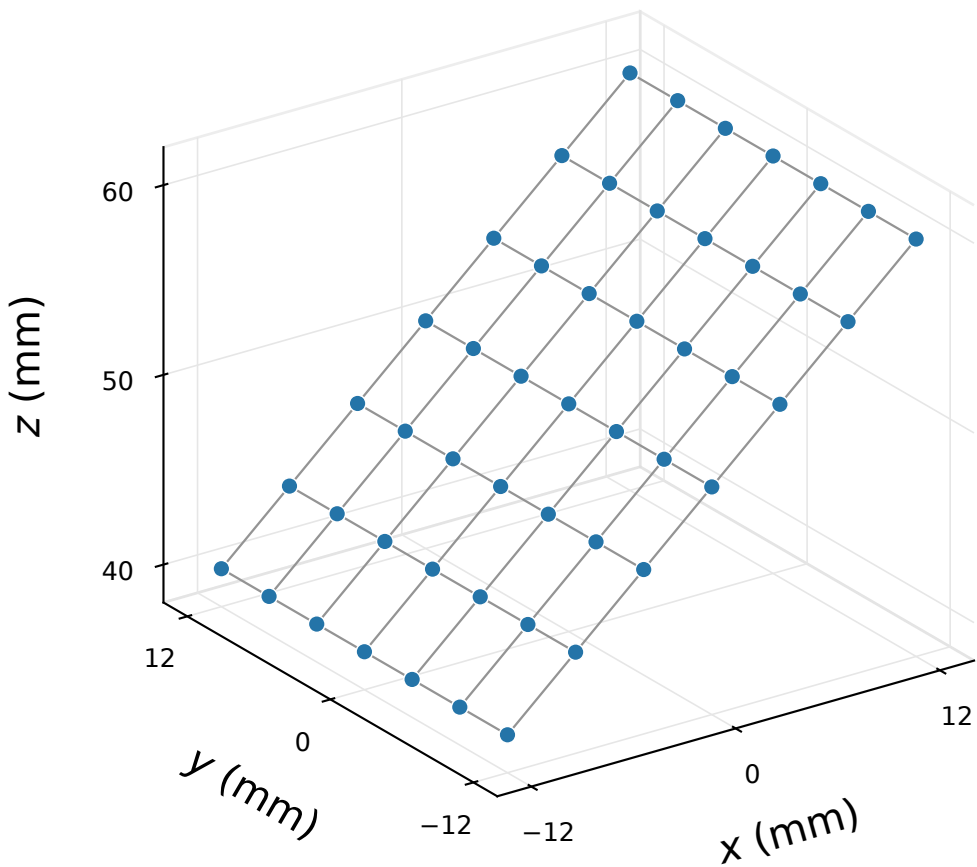
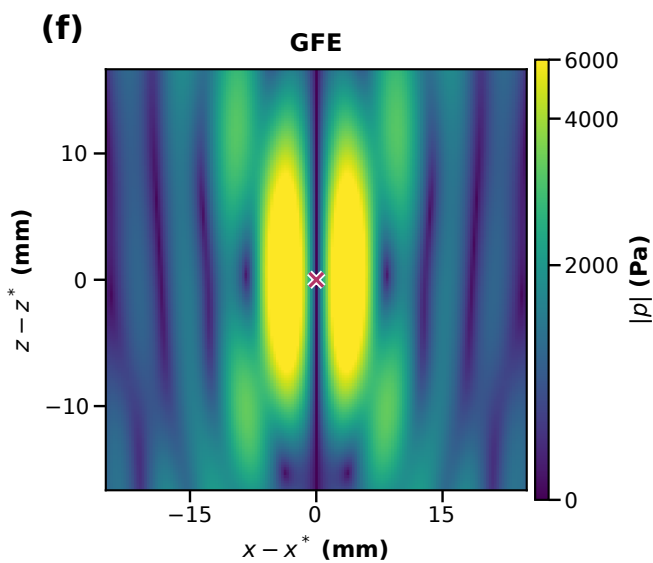
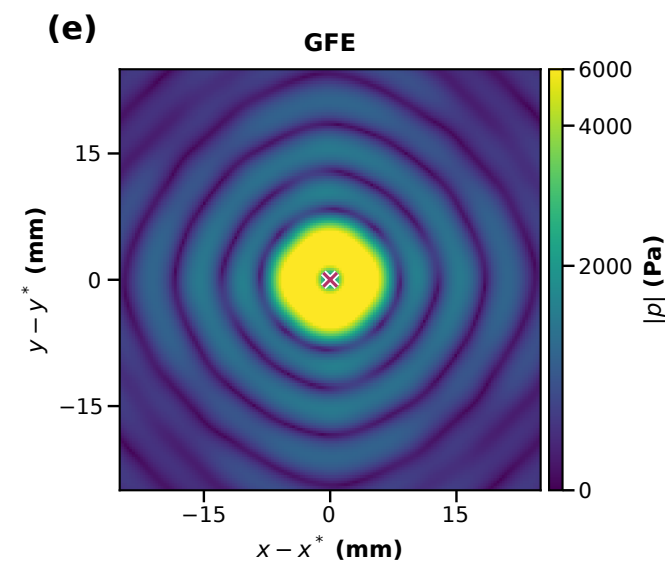
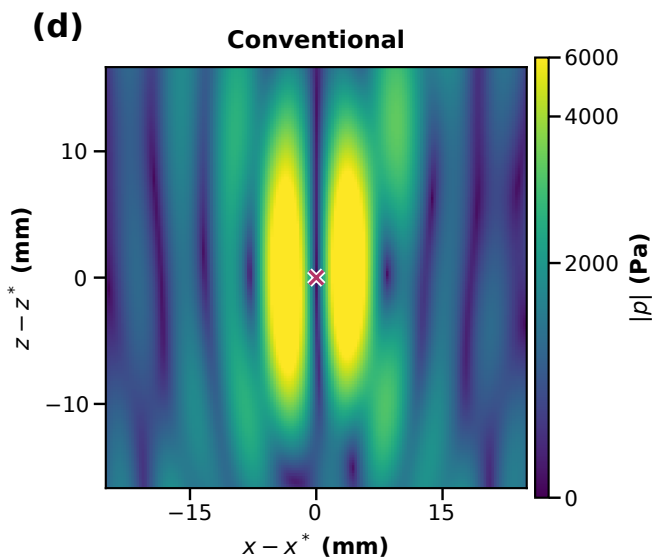
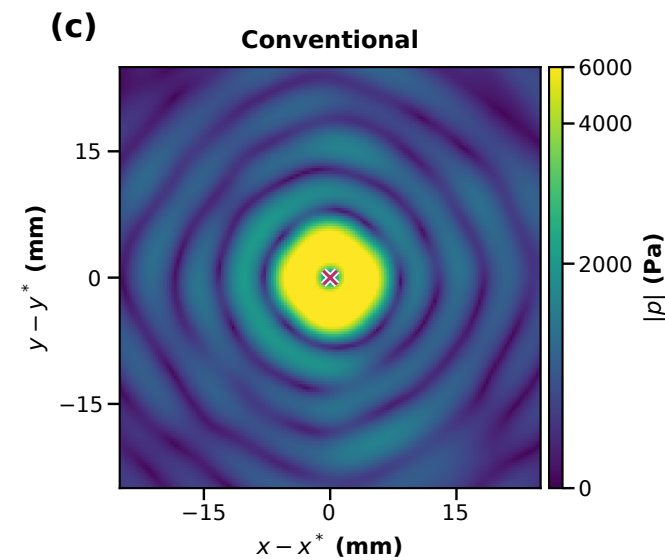
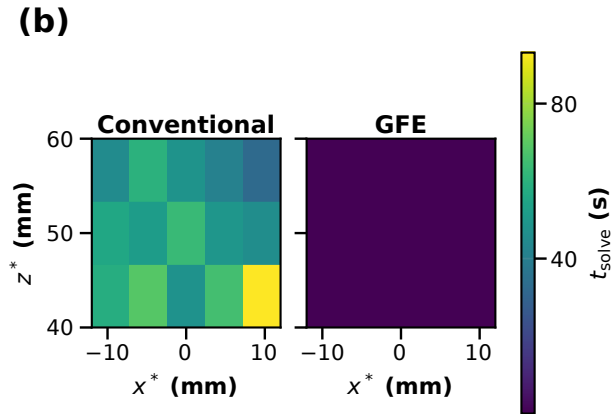
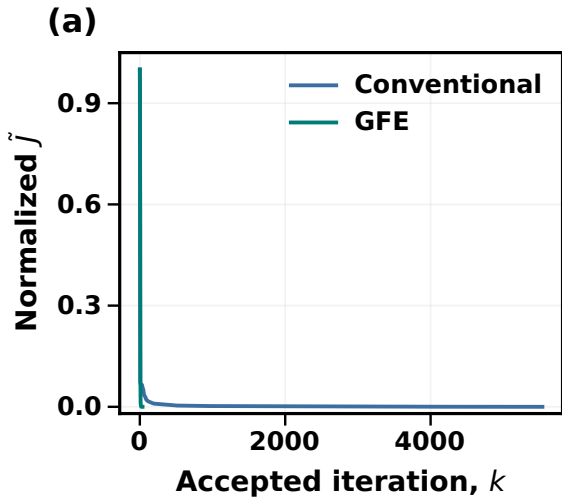
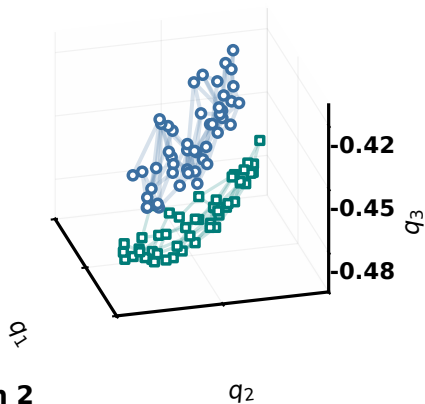


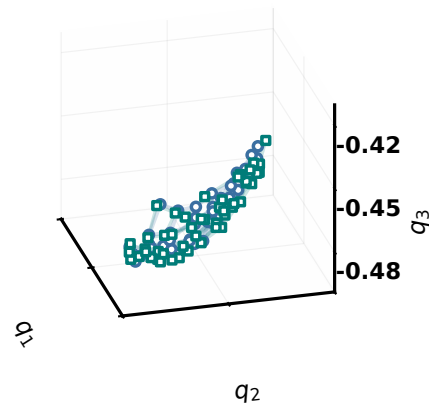




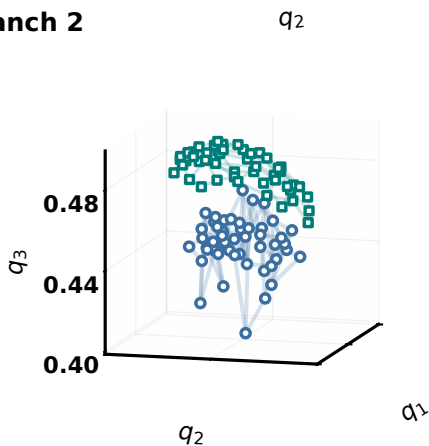
(a) Branch 1 500 iterations



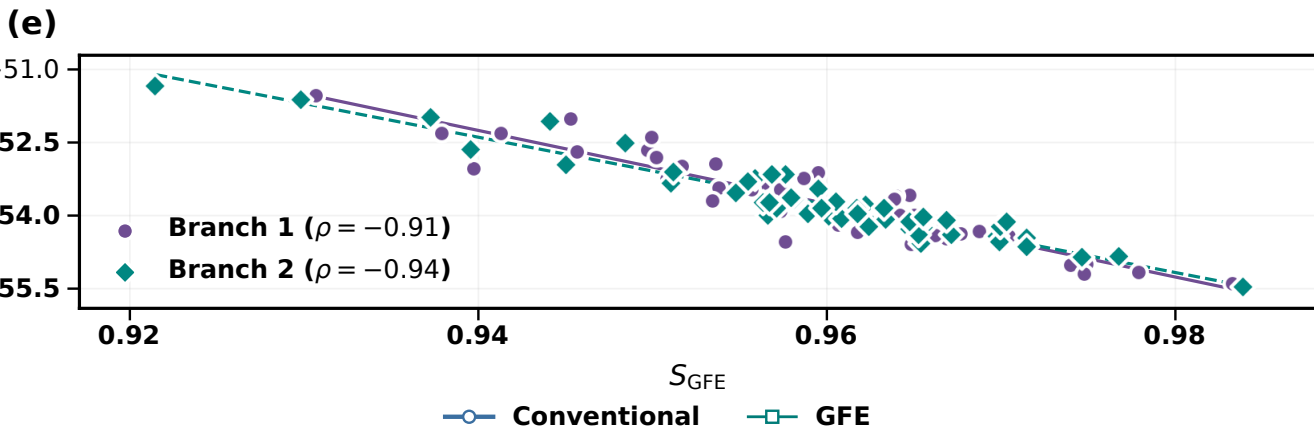
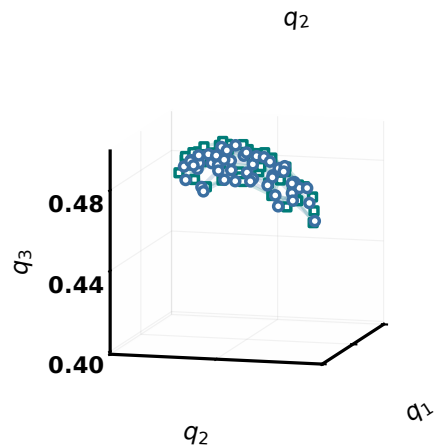
(b) 10,000 iterations

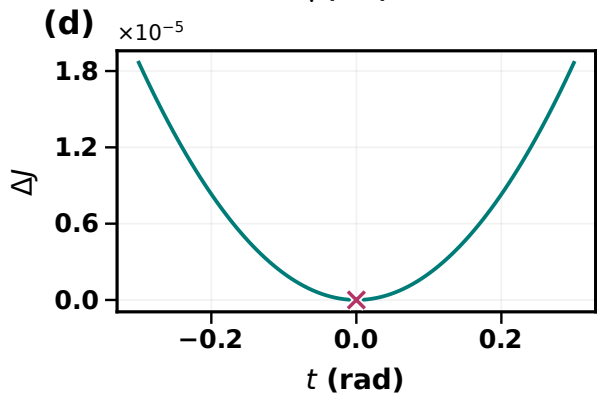
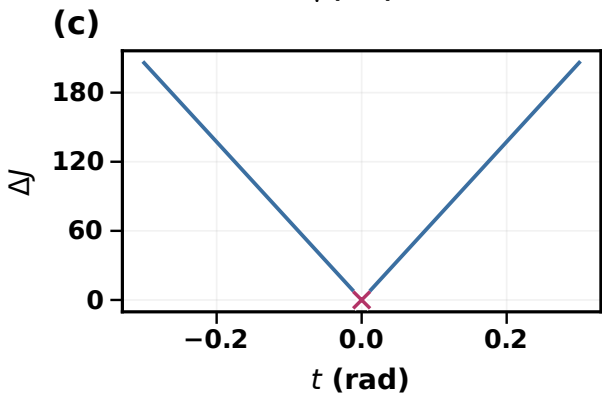
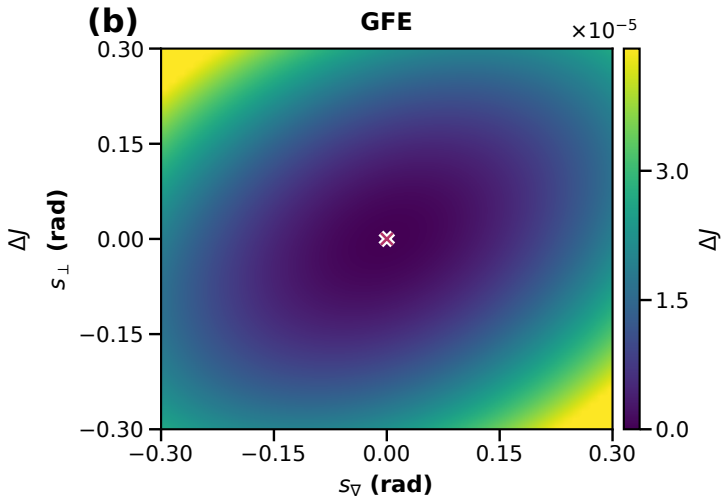
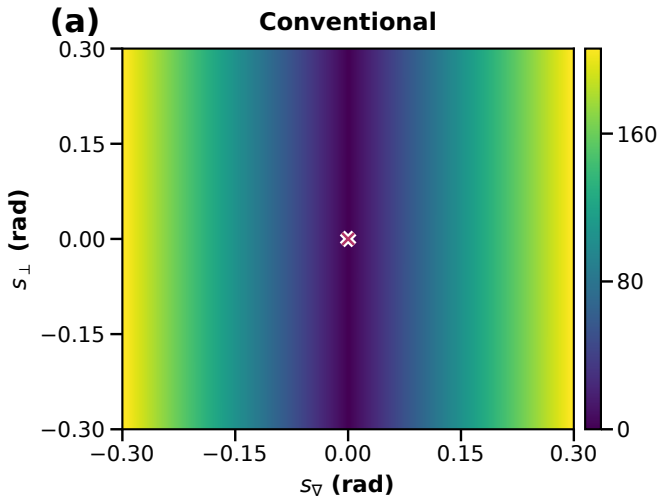


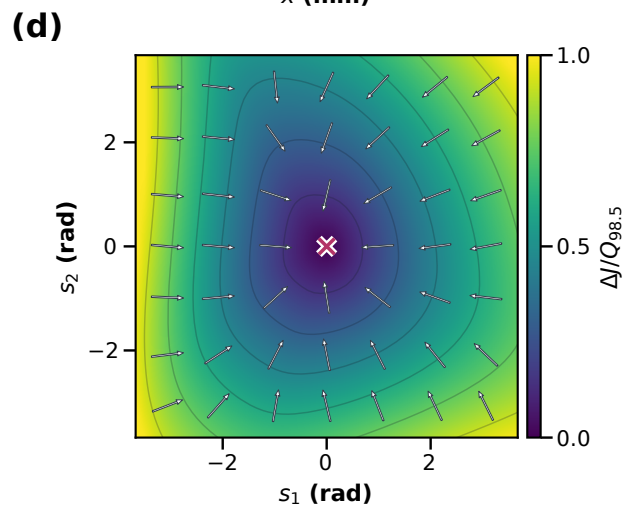
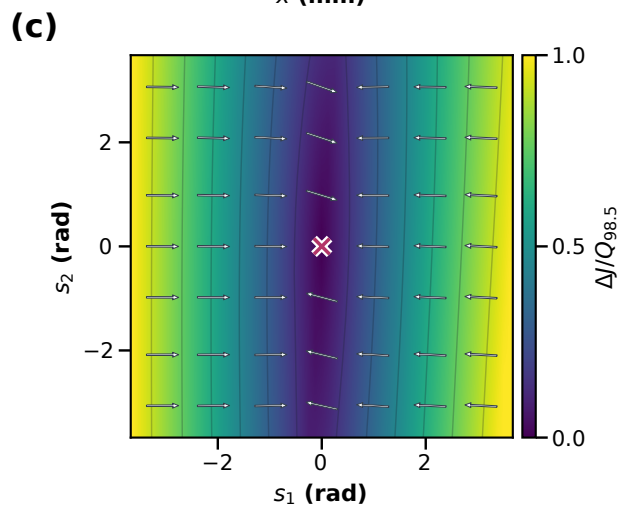
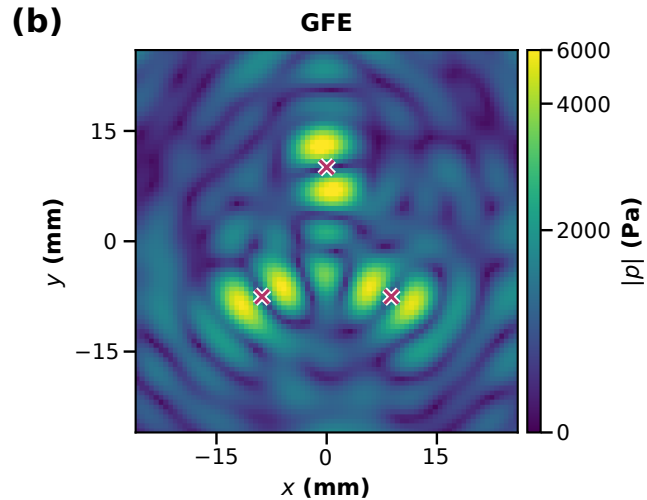
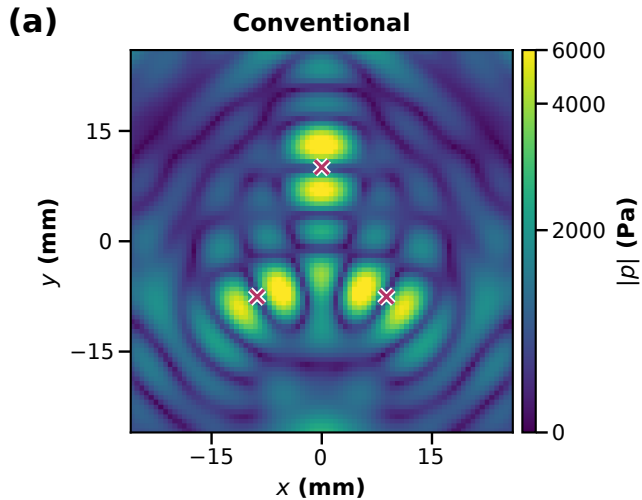
(c) Branch 2

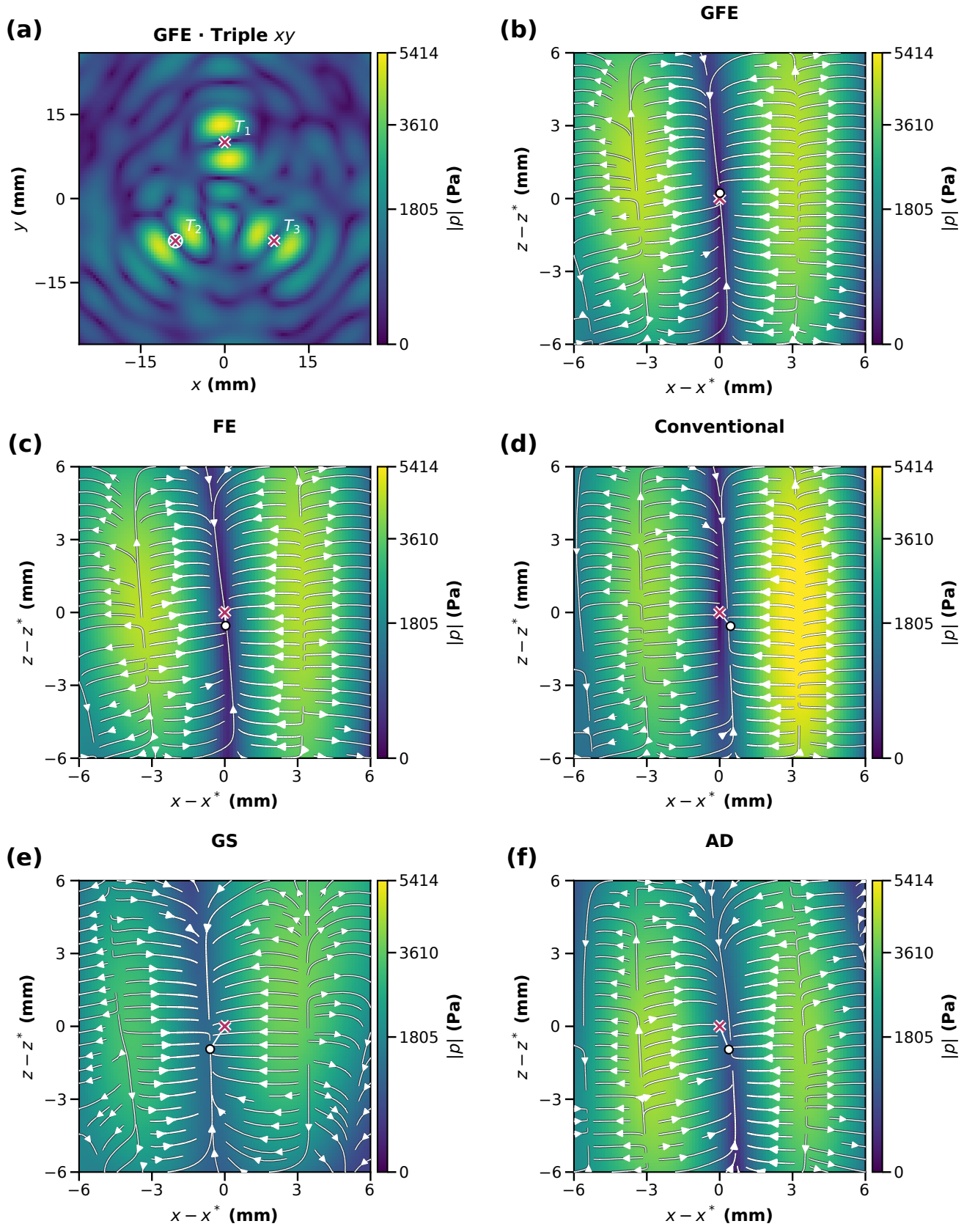


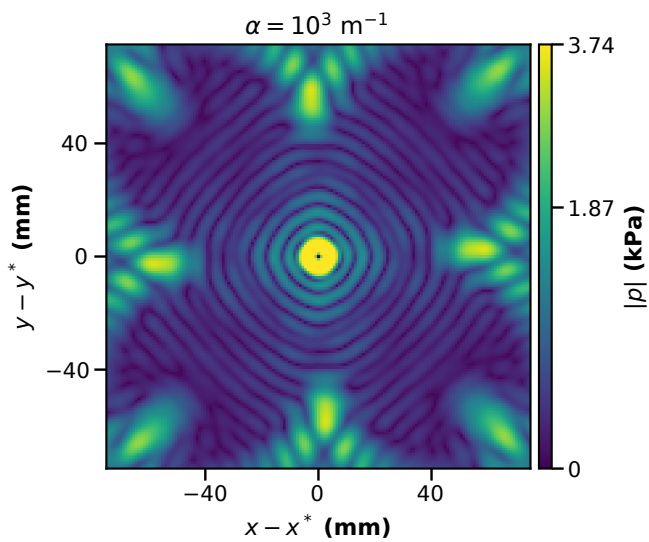
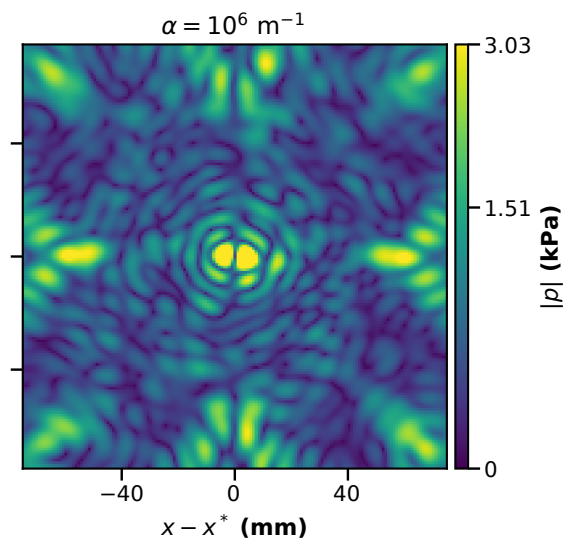
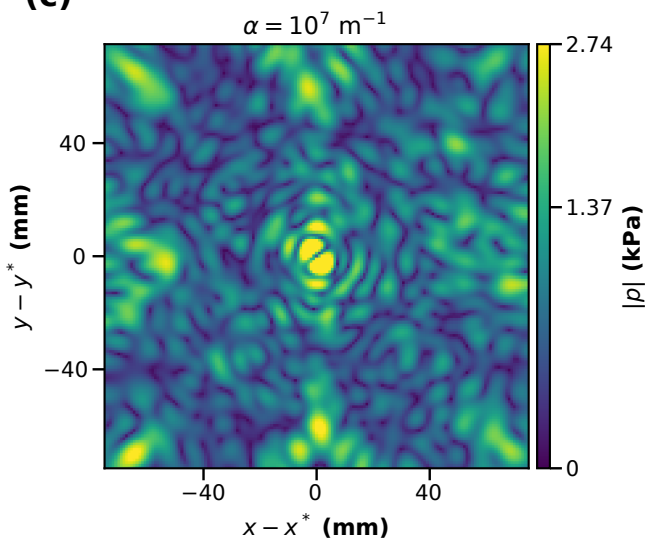
(d)





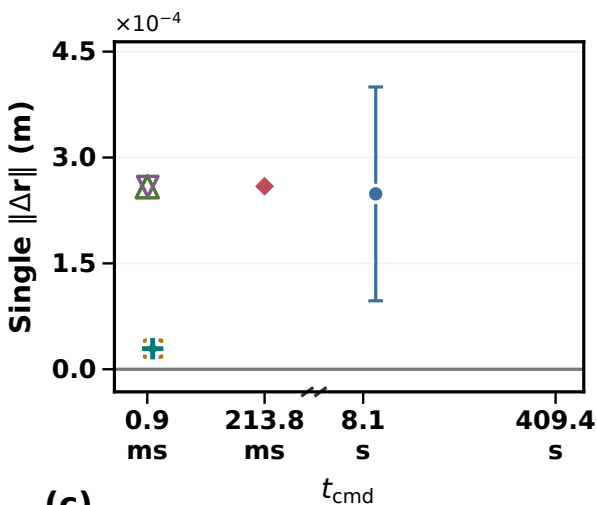




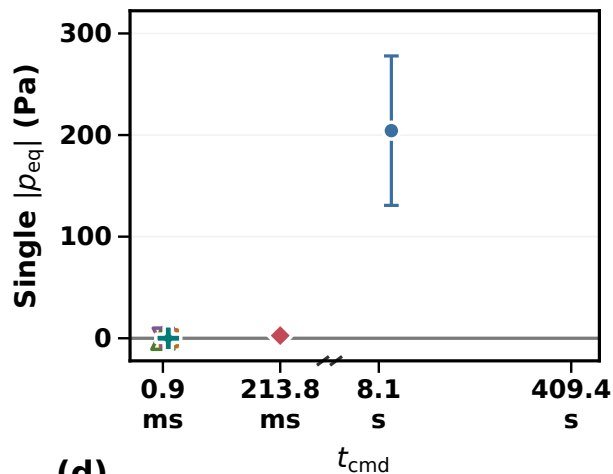
(a)**(b)****(c)**

+ GFE • Conventional ▽ GS
 □ FE △ IB ♦ AD

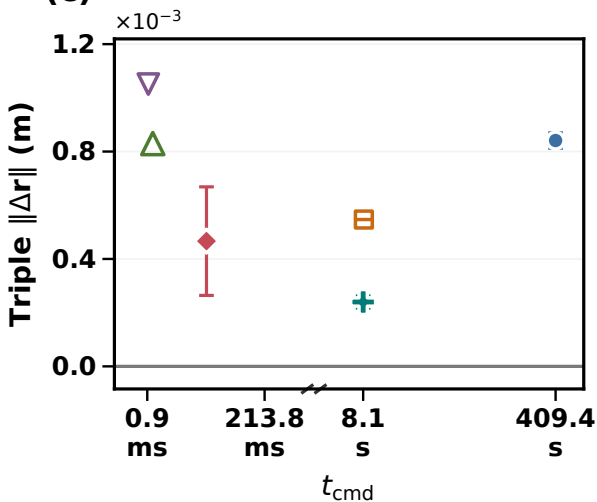
(a)



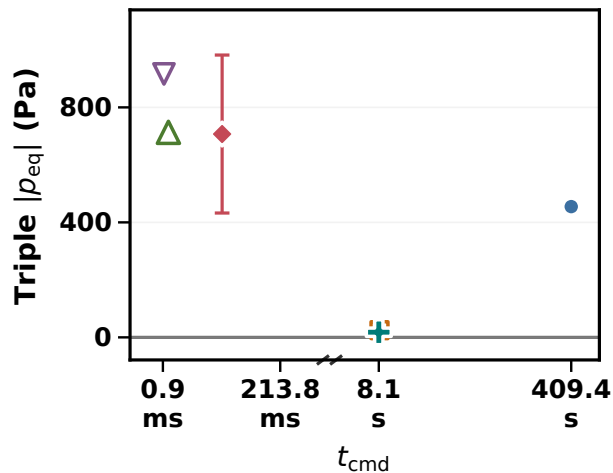
(b)

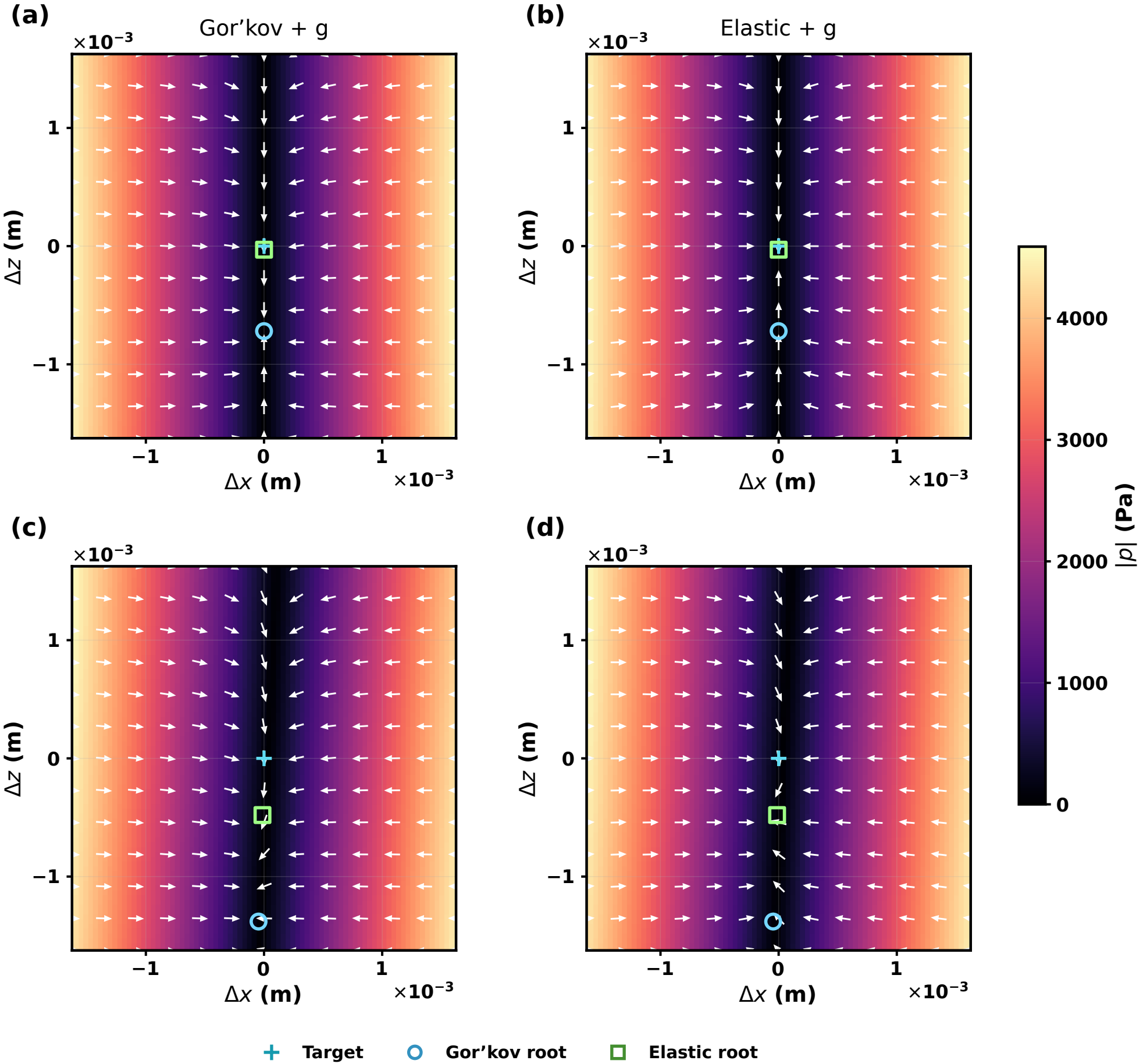


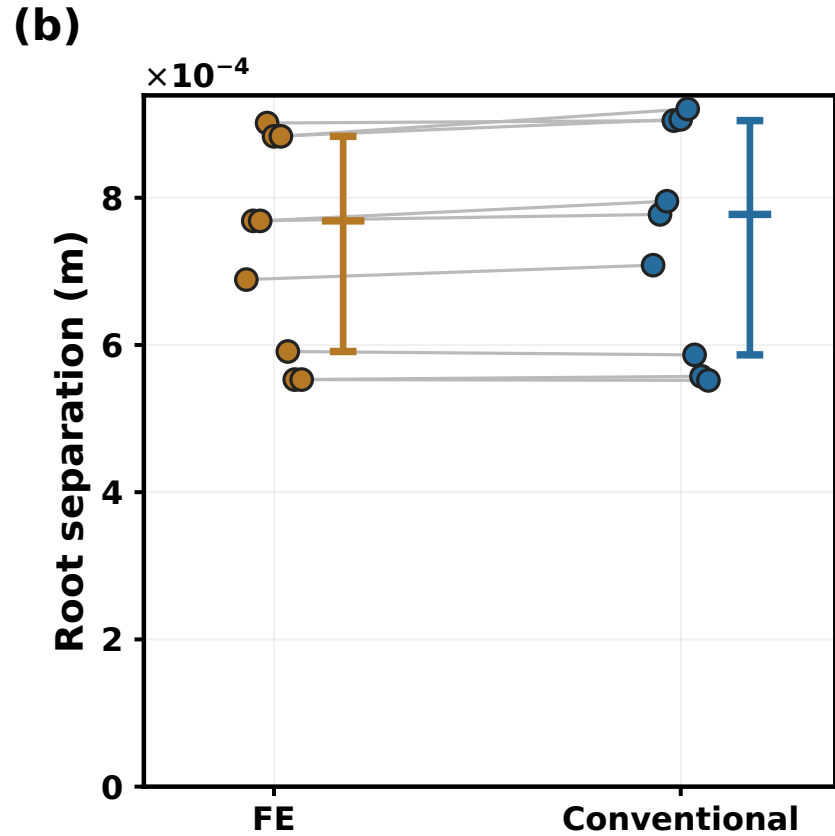
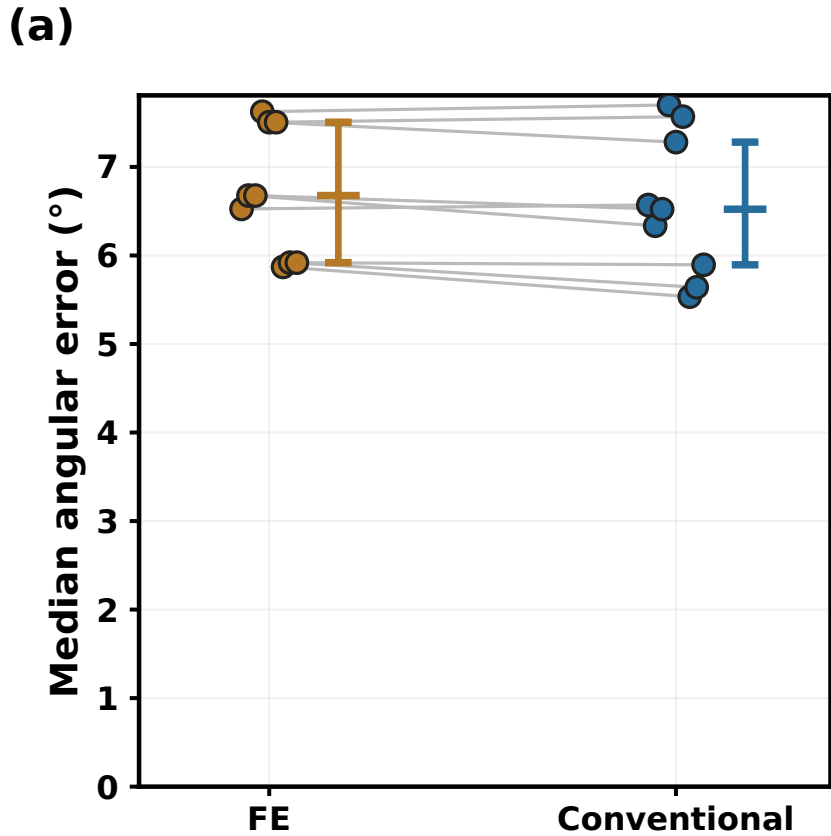
(c)

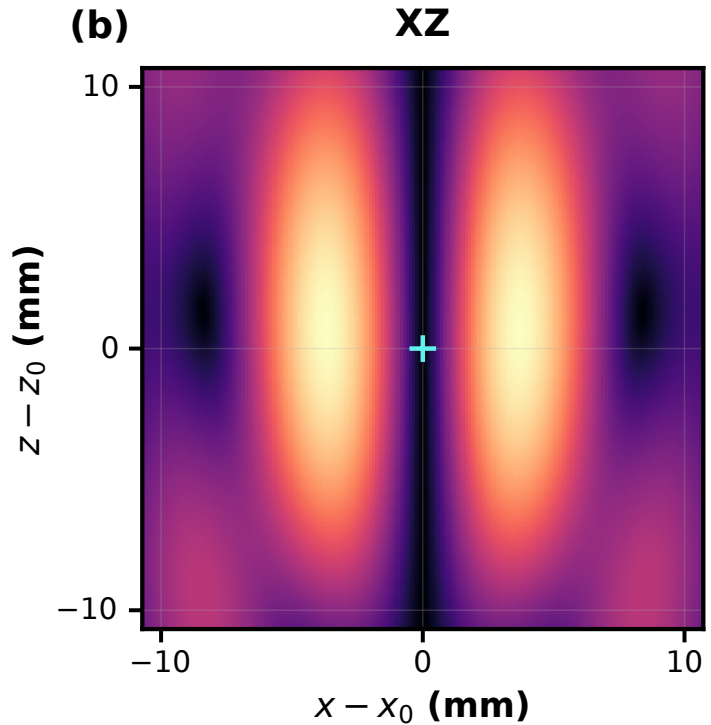
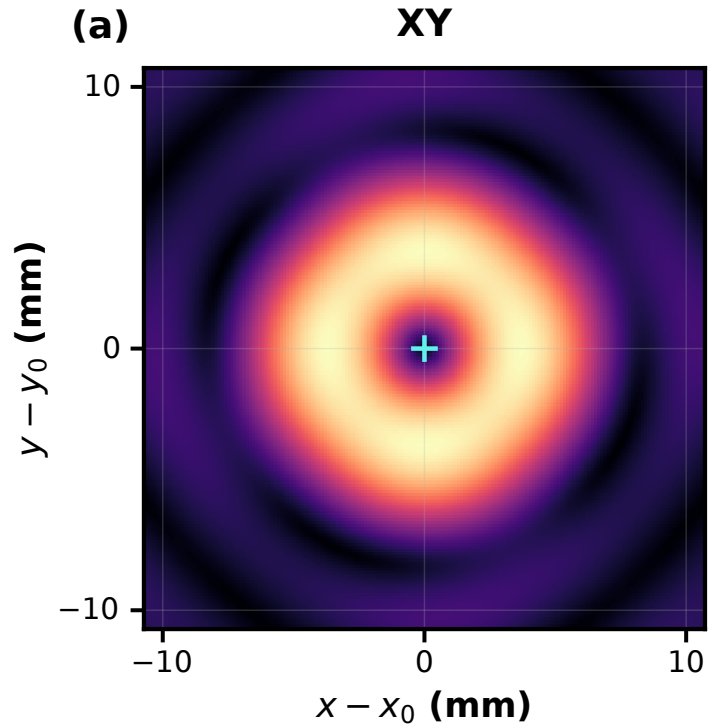
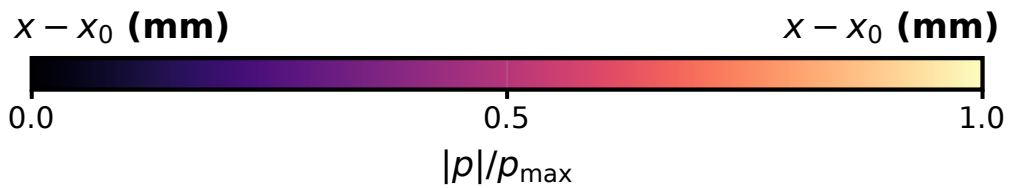
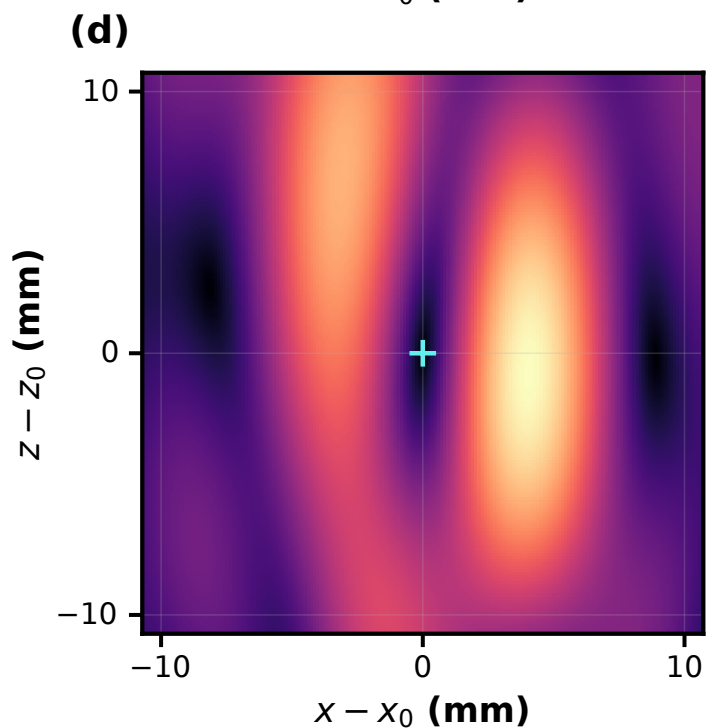
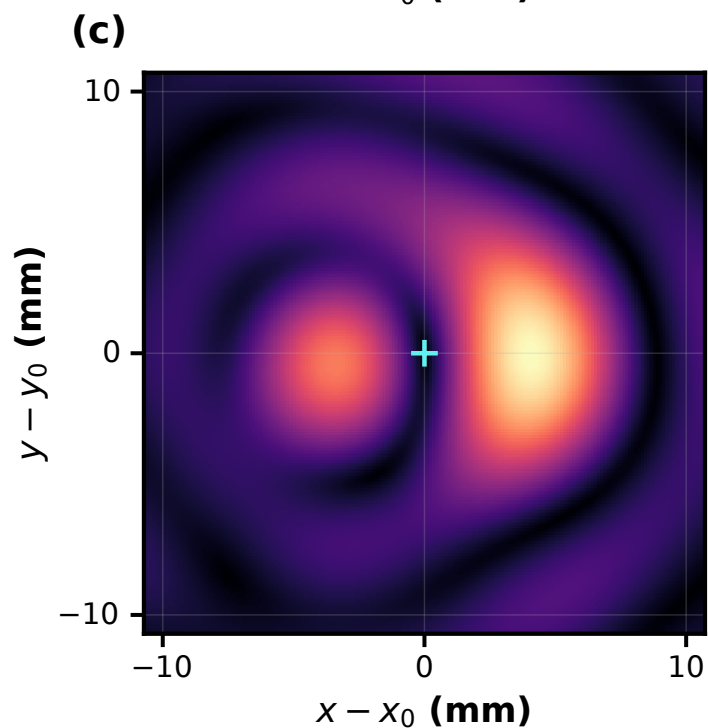


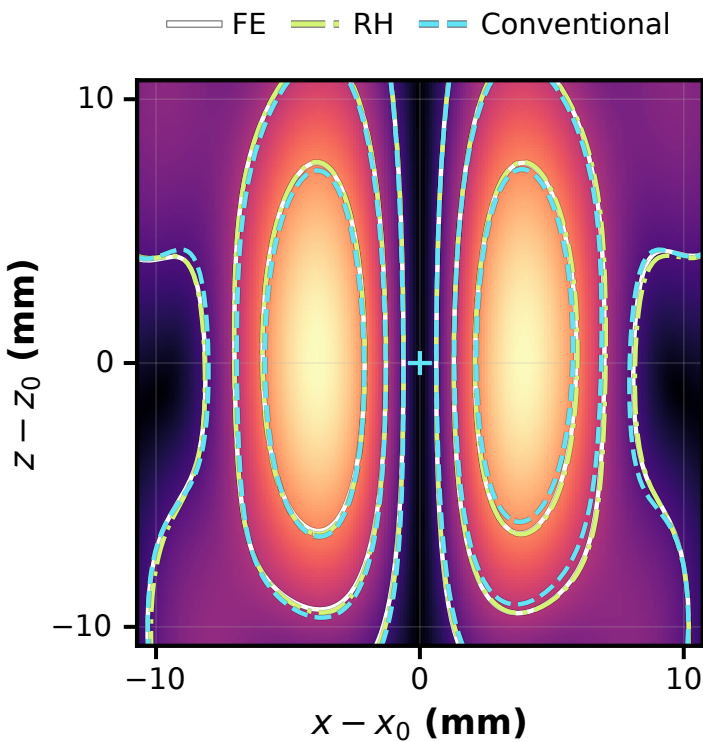
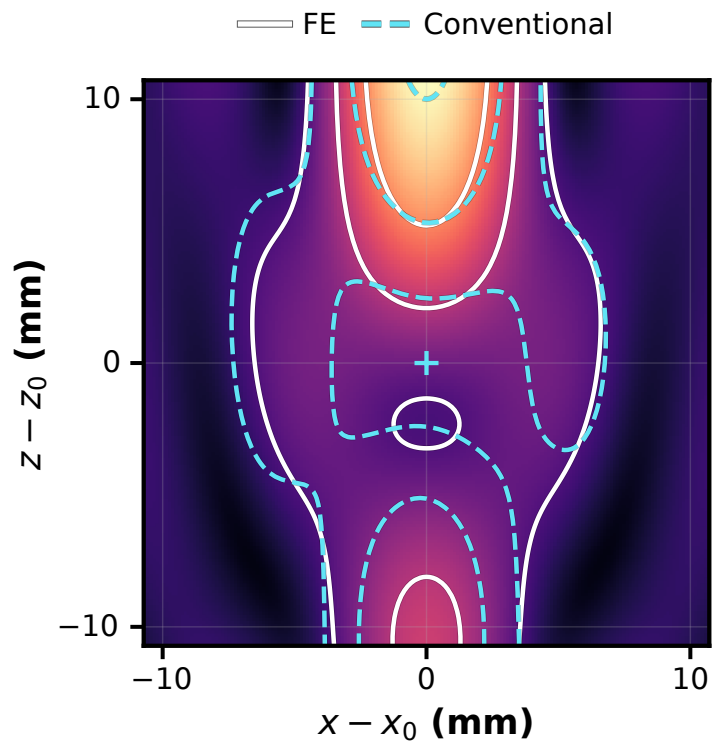
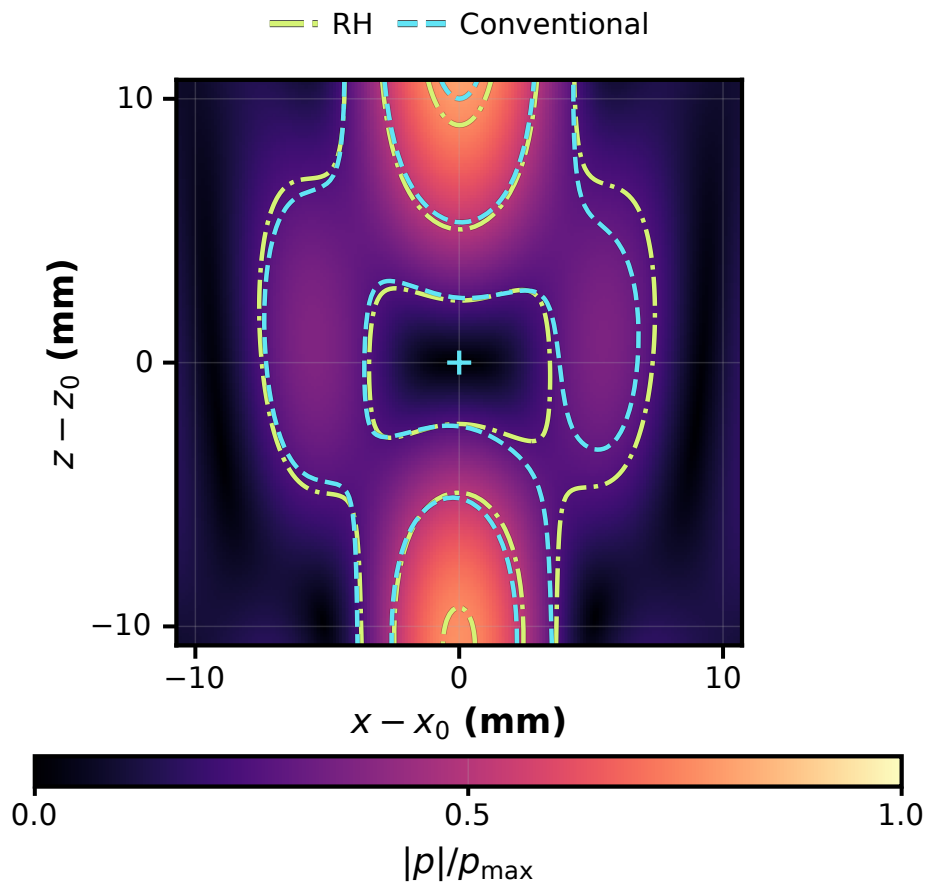
(d)

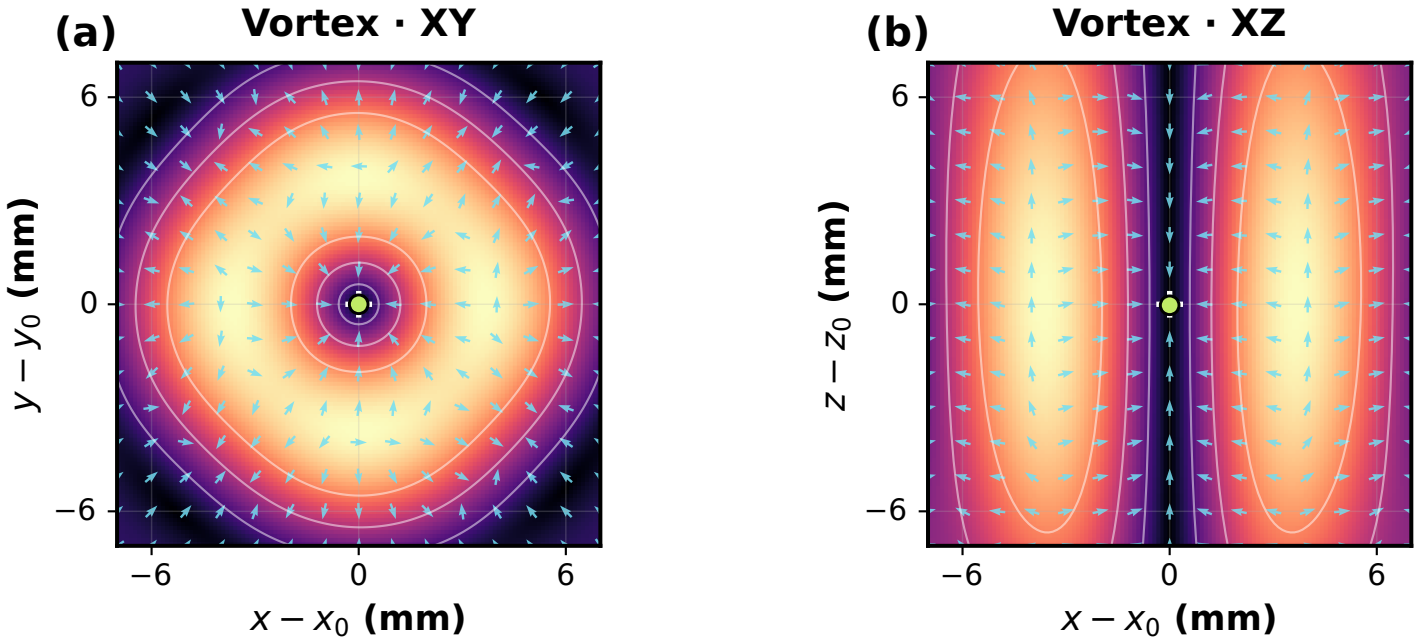




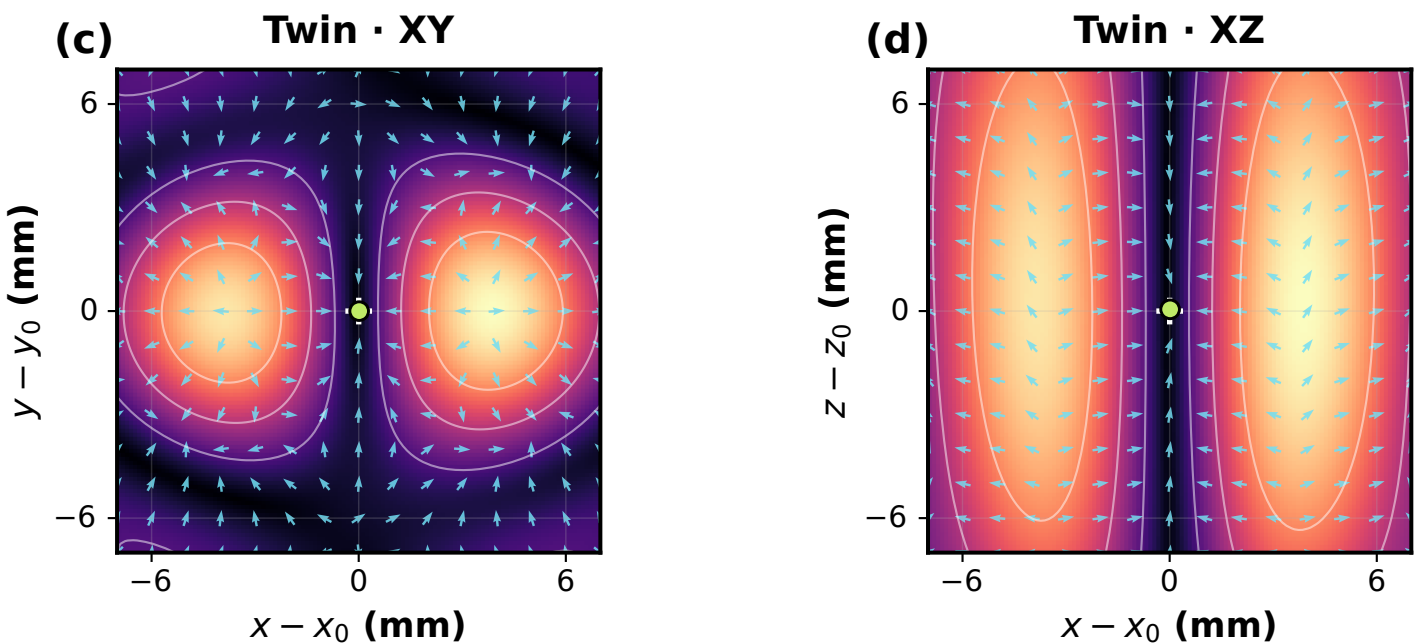


FE**Conventional**

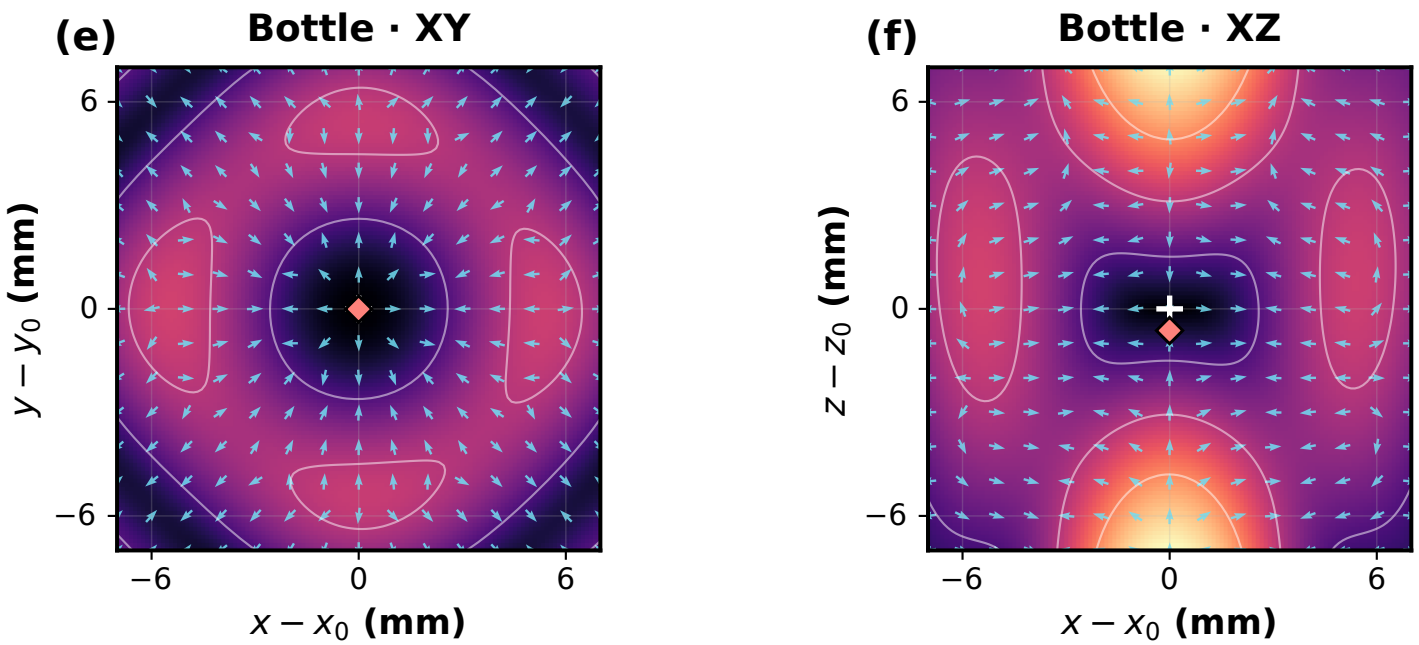
(a) Twin**(b) Bottle****(c) Bottle**



GFE · standard sign

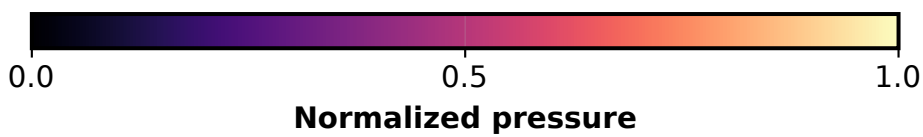


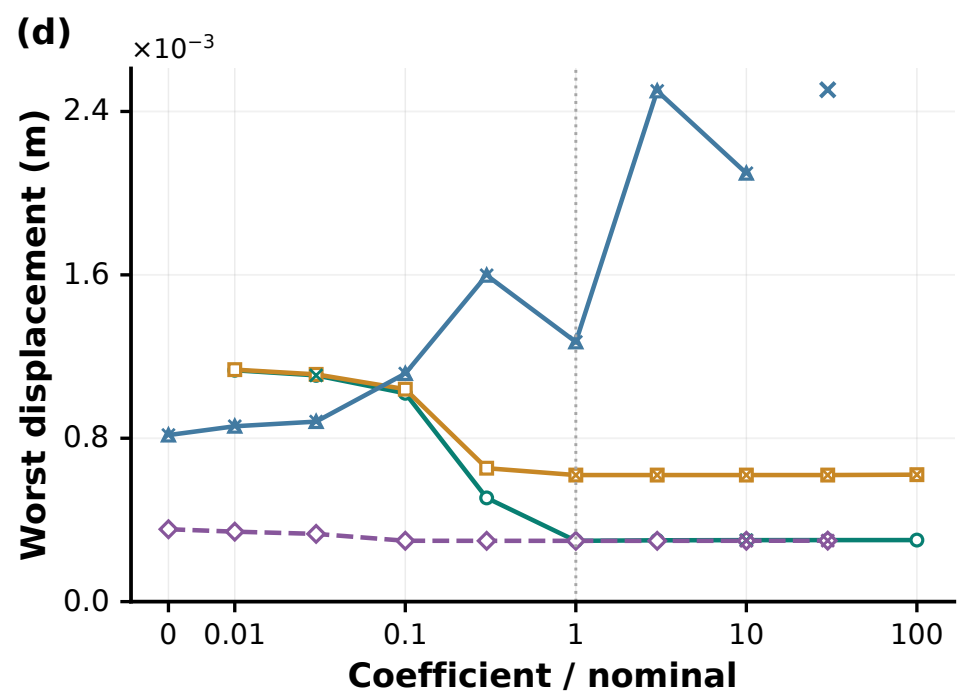
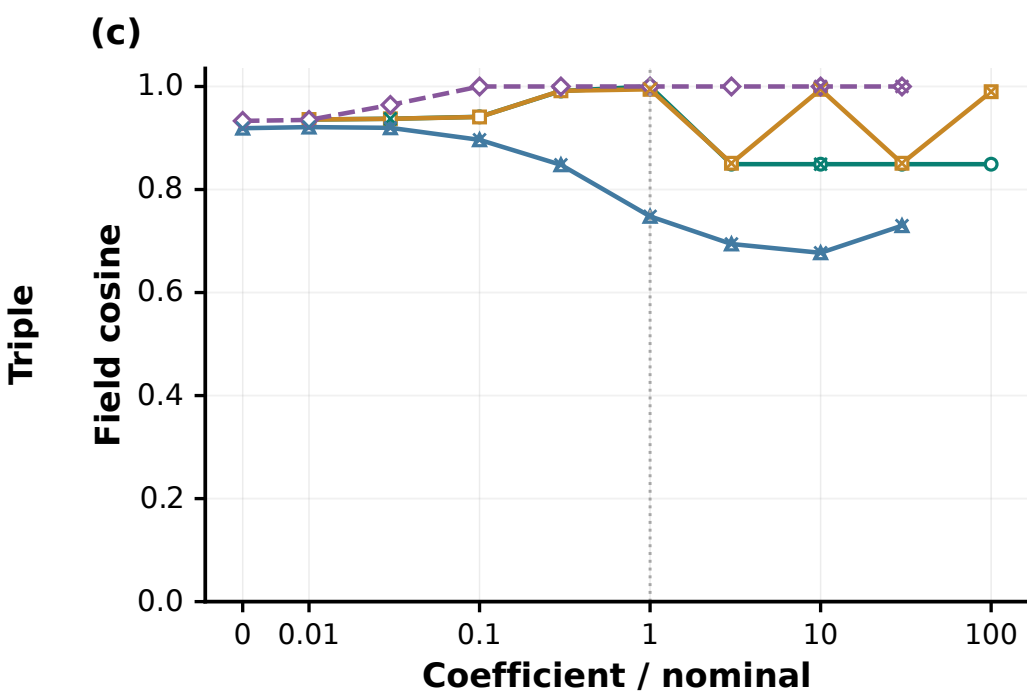
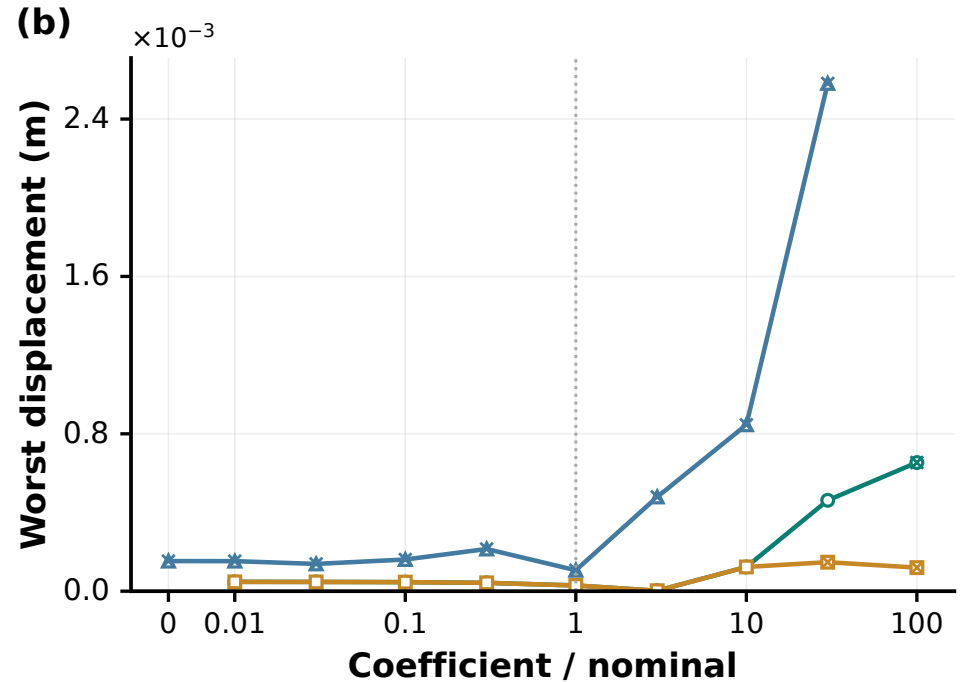
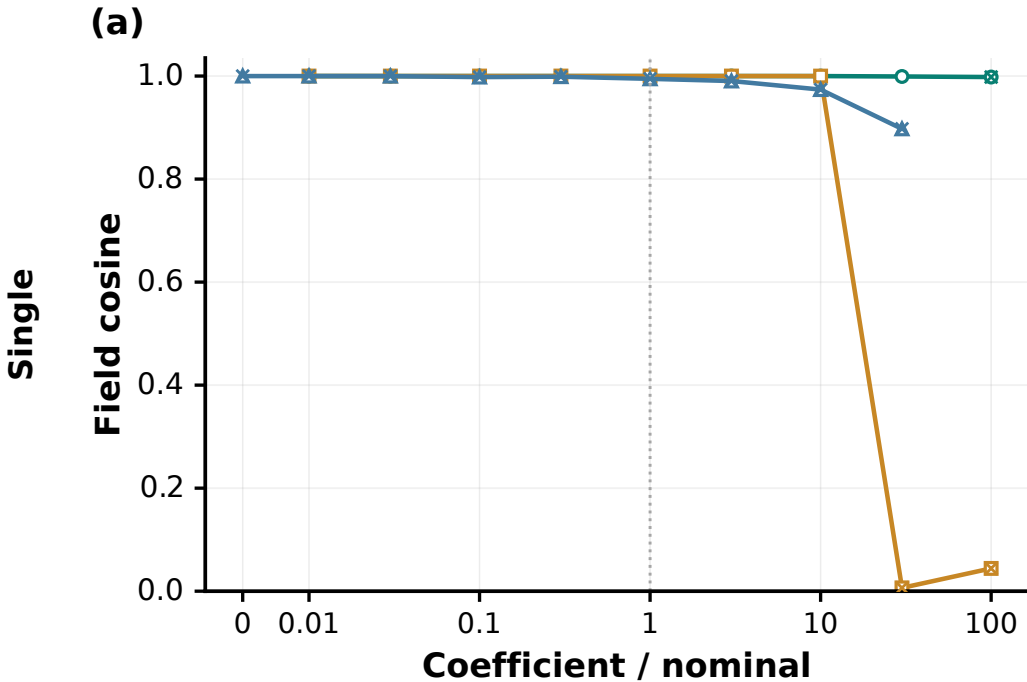
FE · standard sign

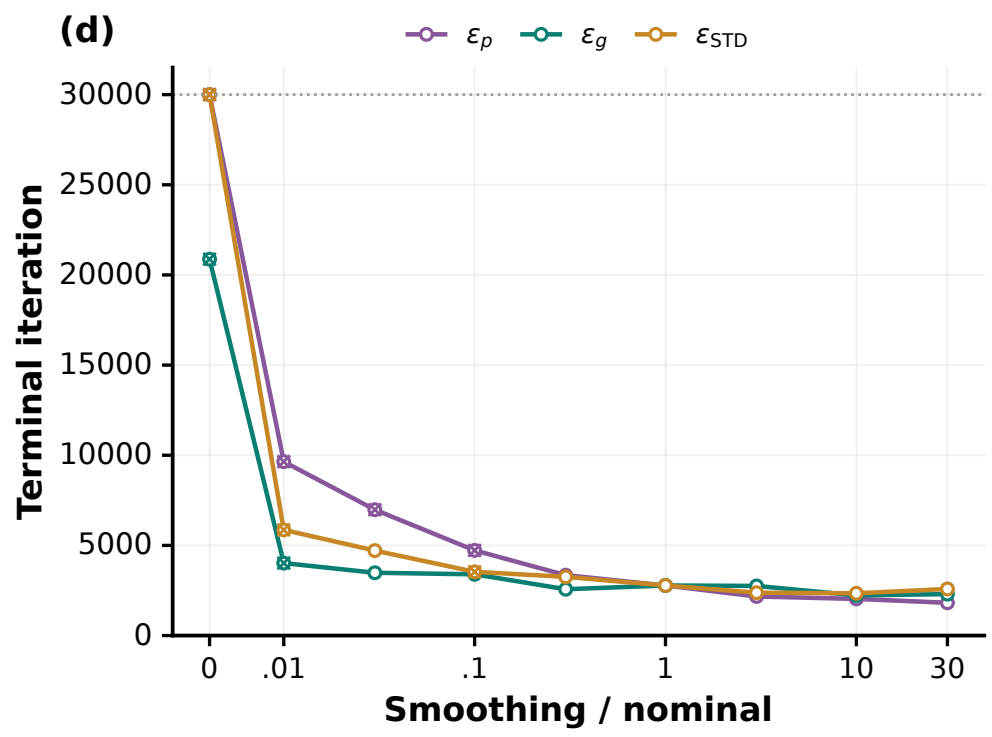
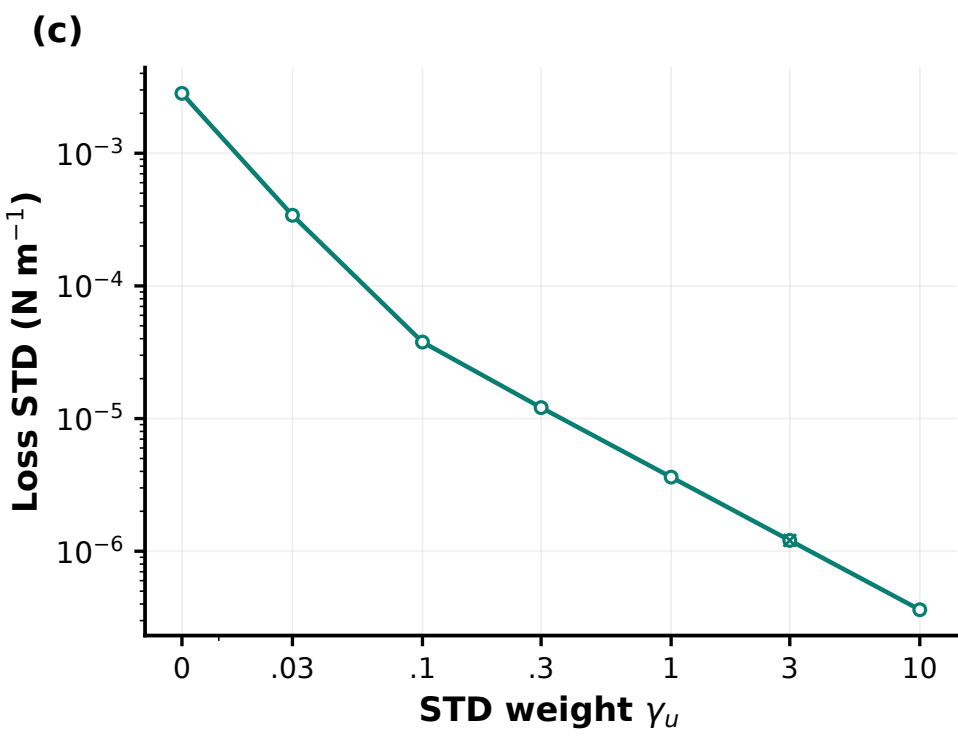
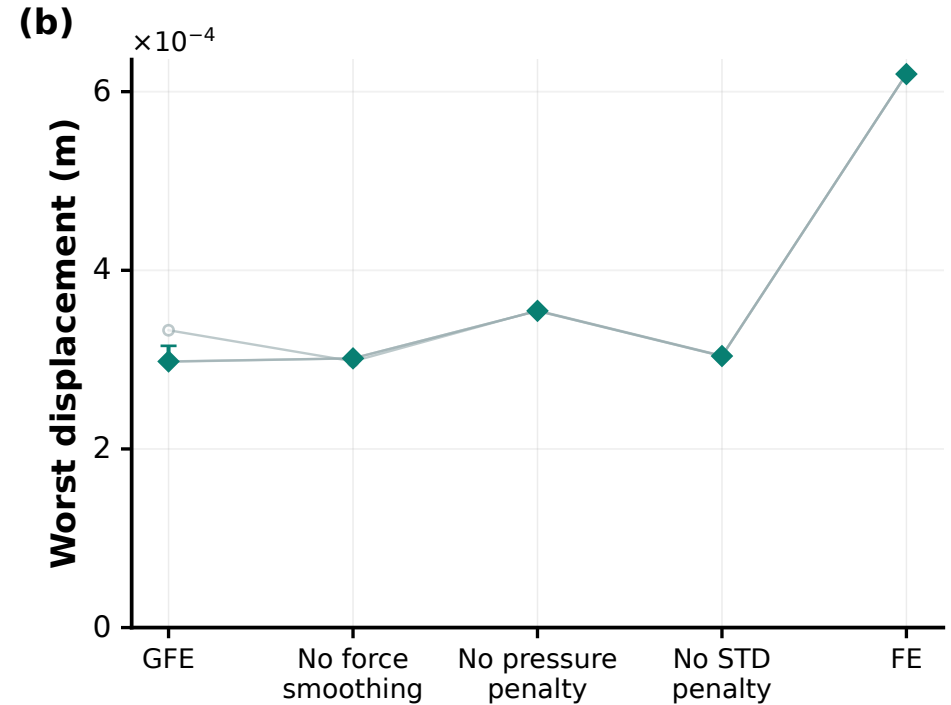
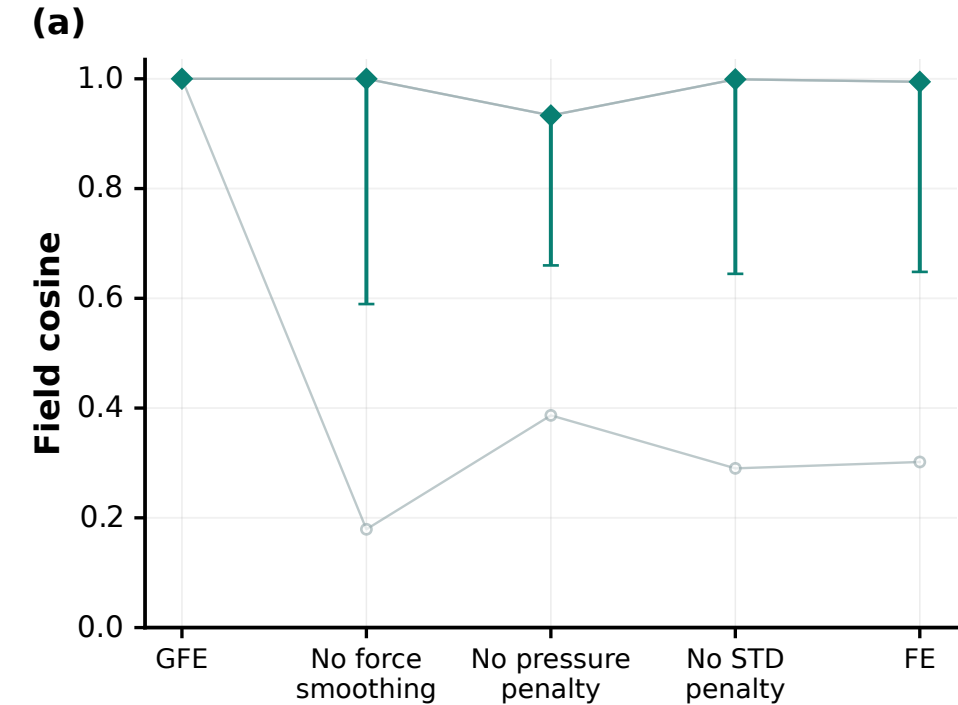


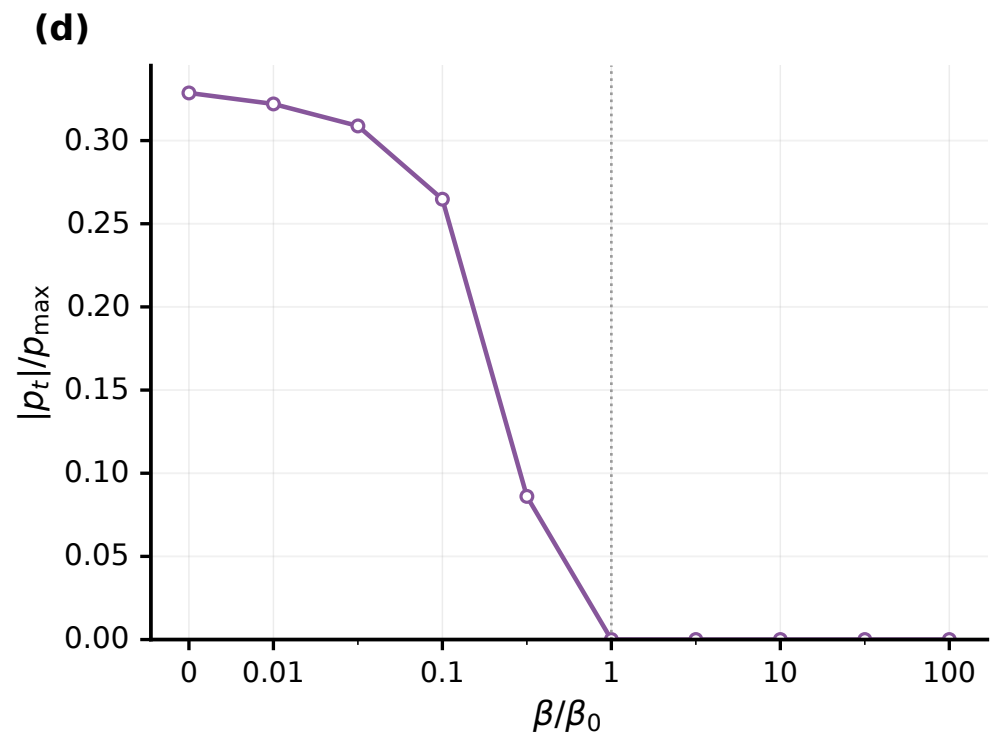
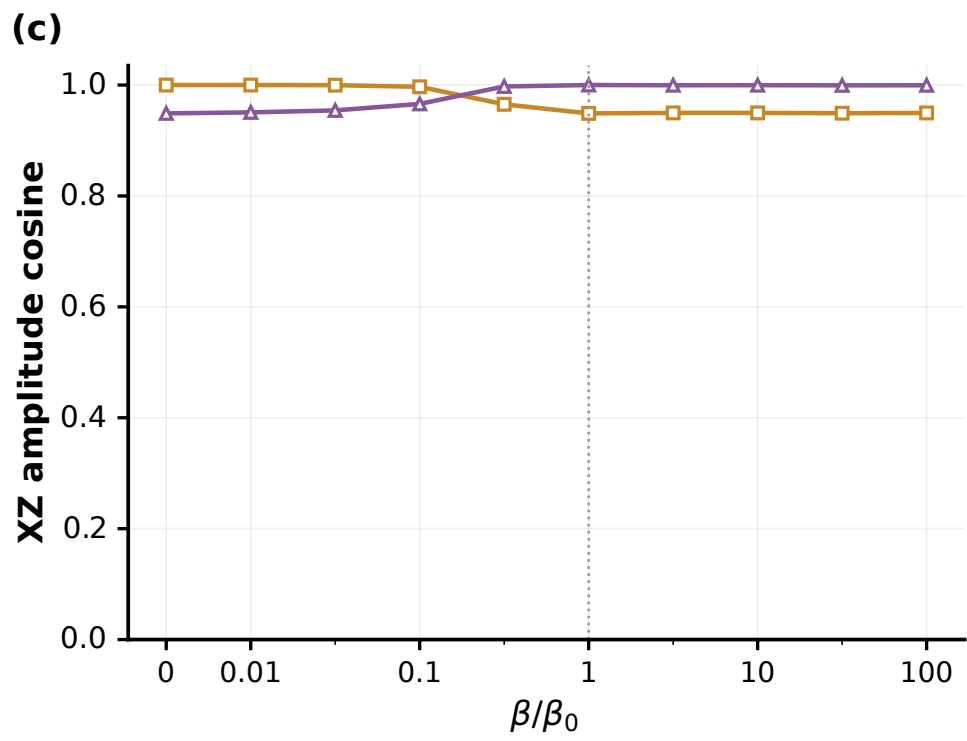
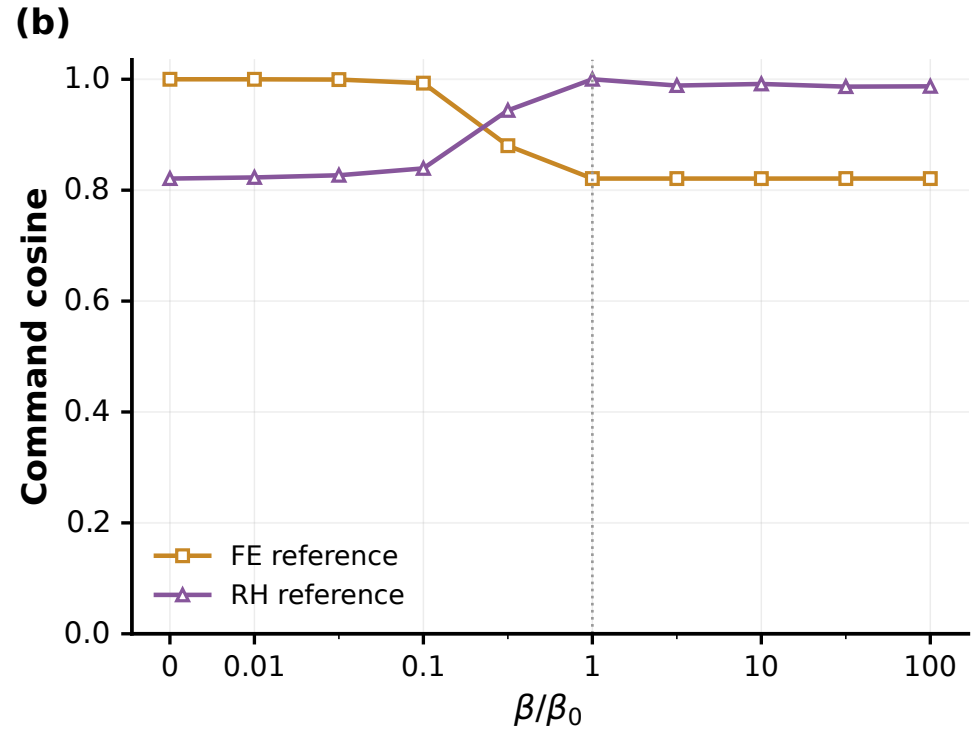
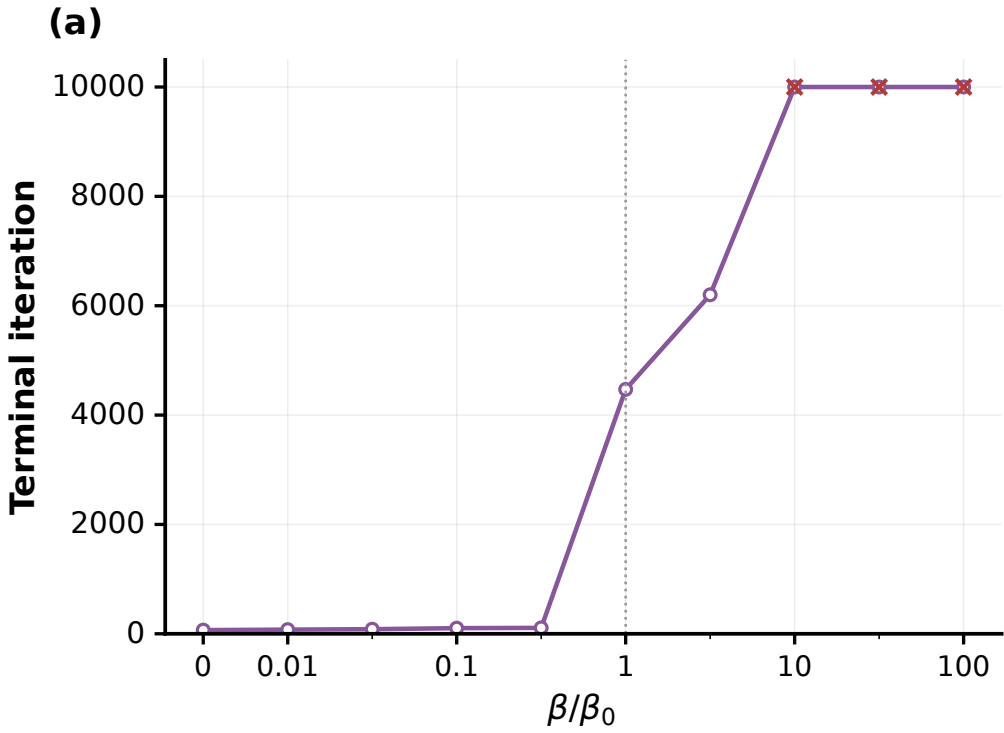
RH · alternative sign

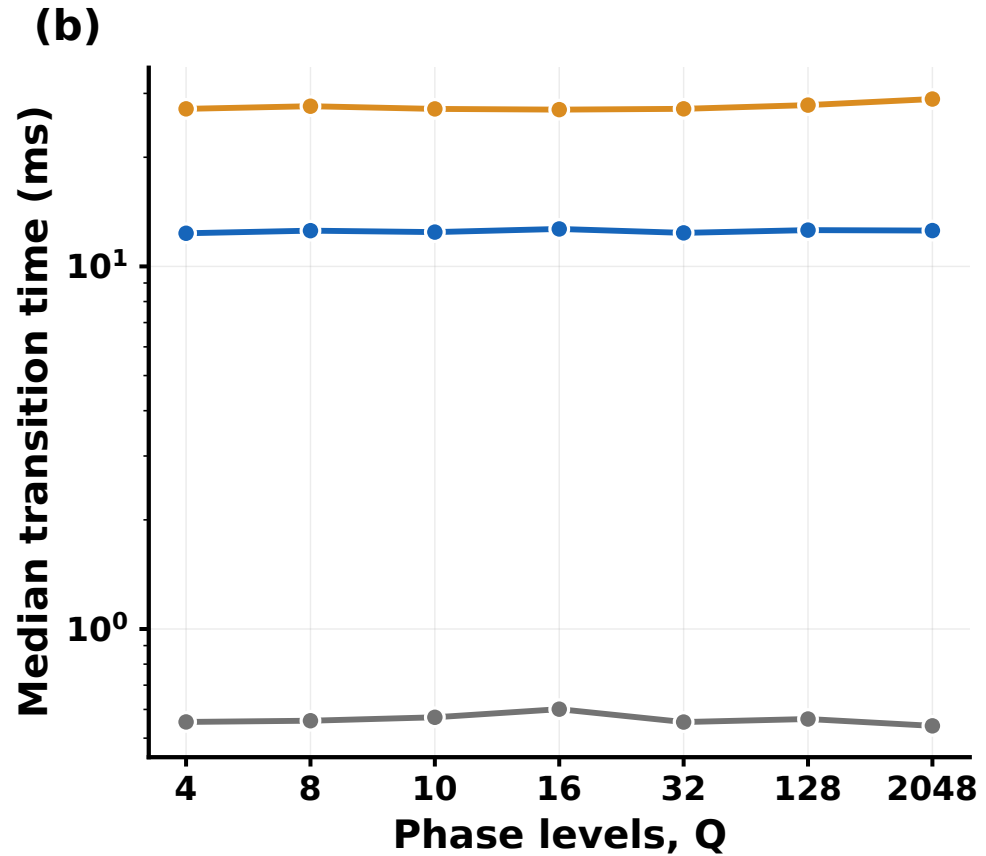
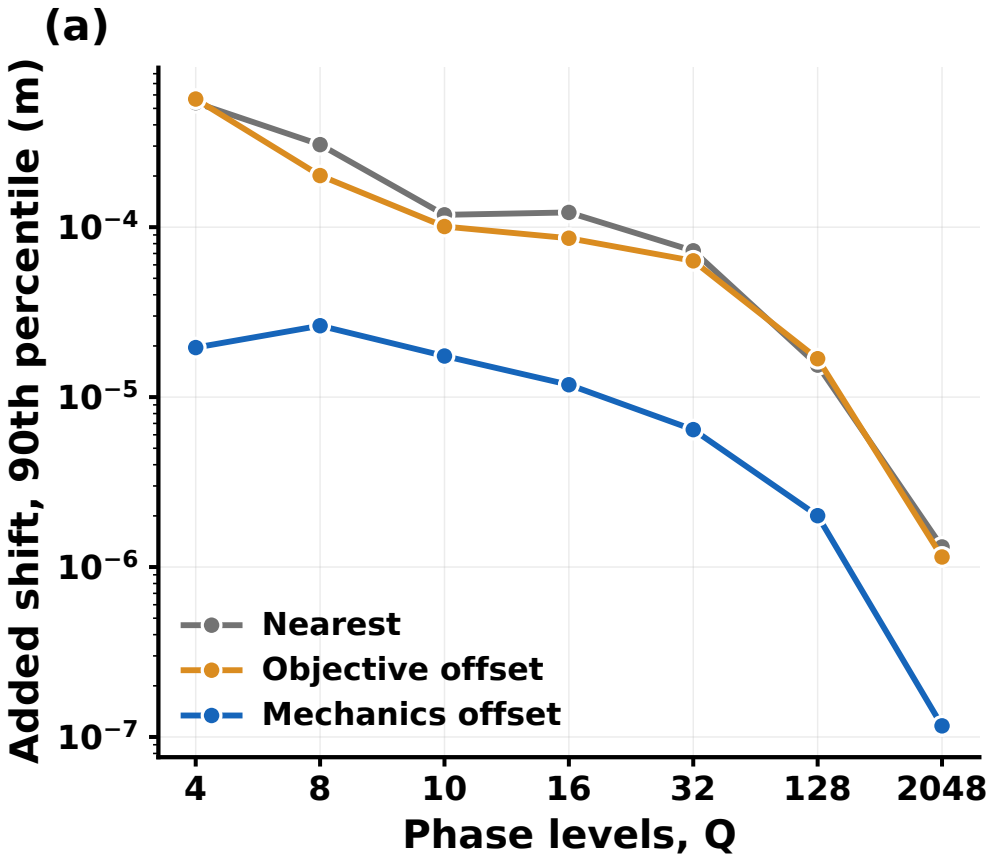
⊕ Target ● Restoring root ◆ Non-restoring root

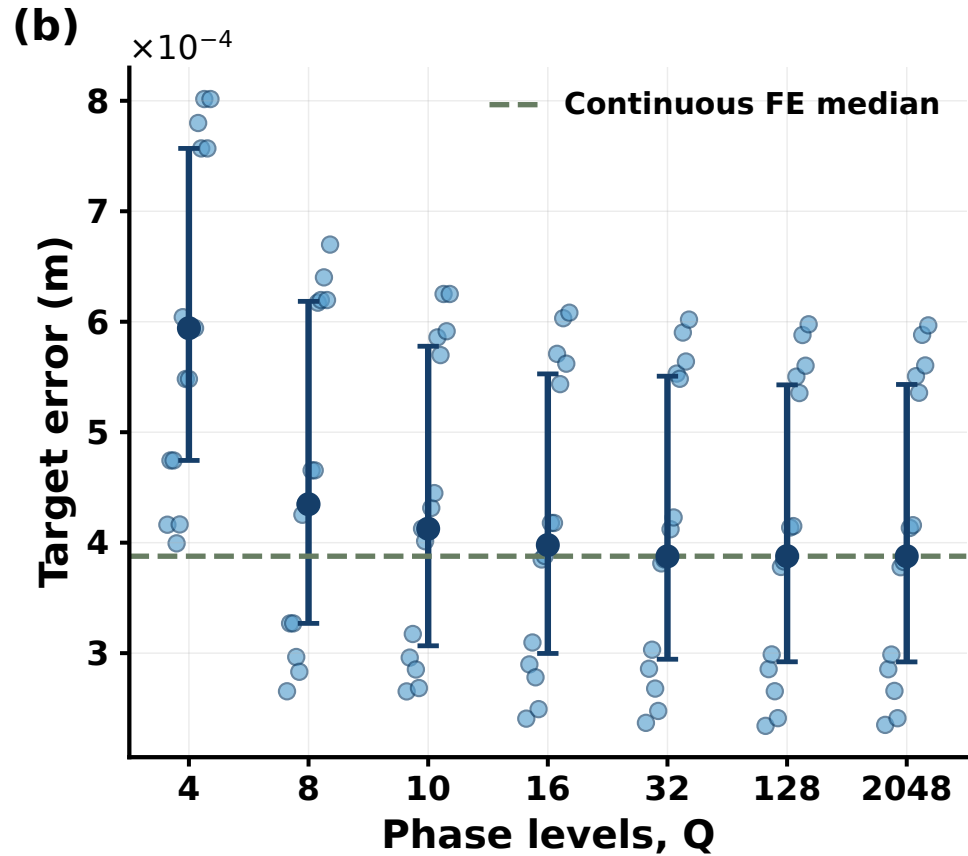
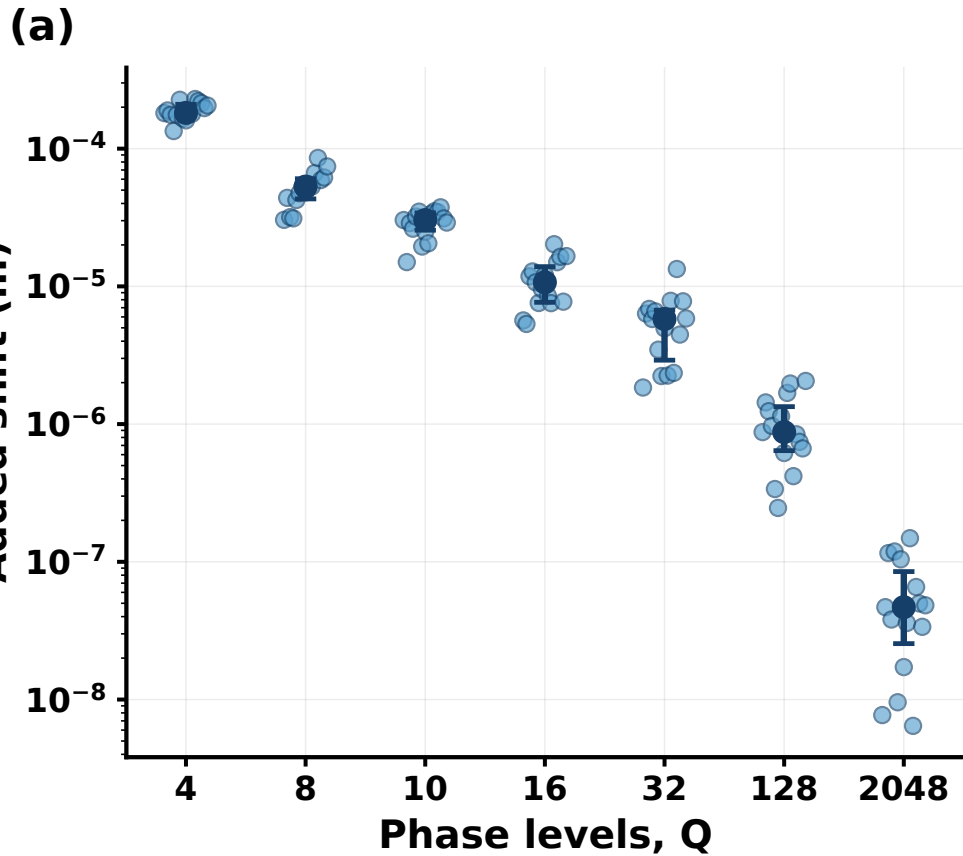


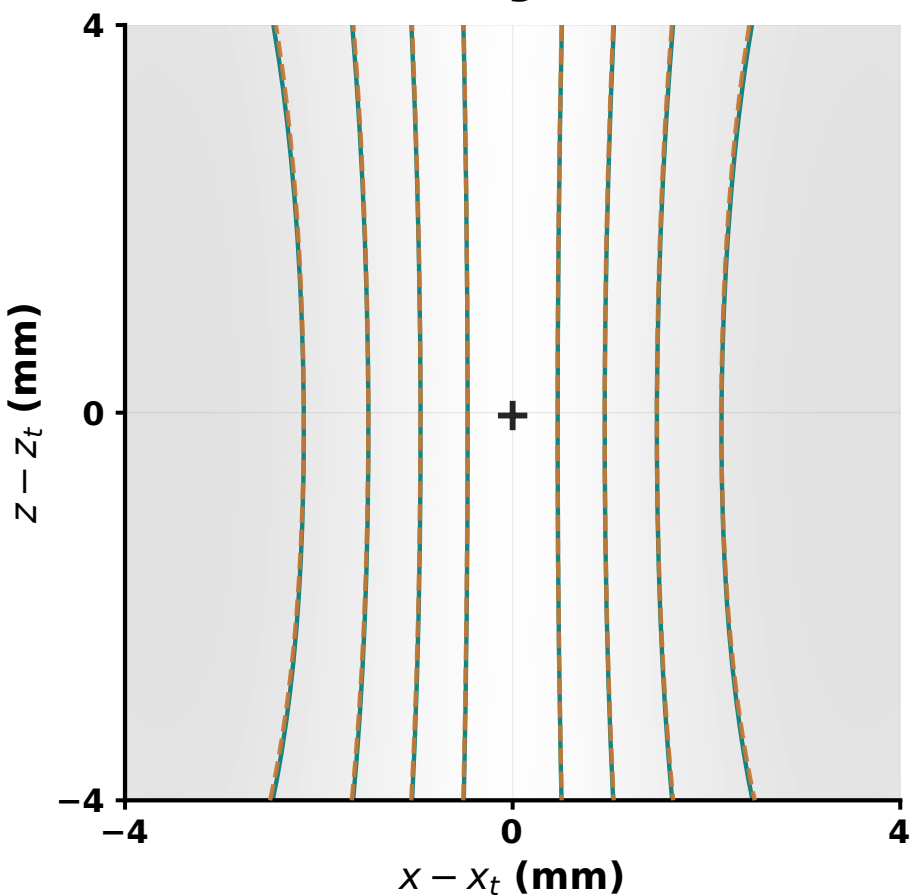
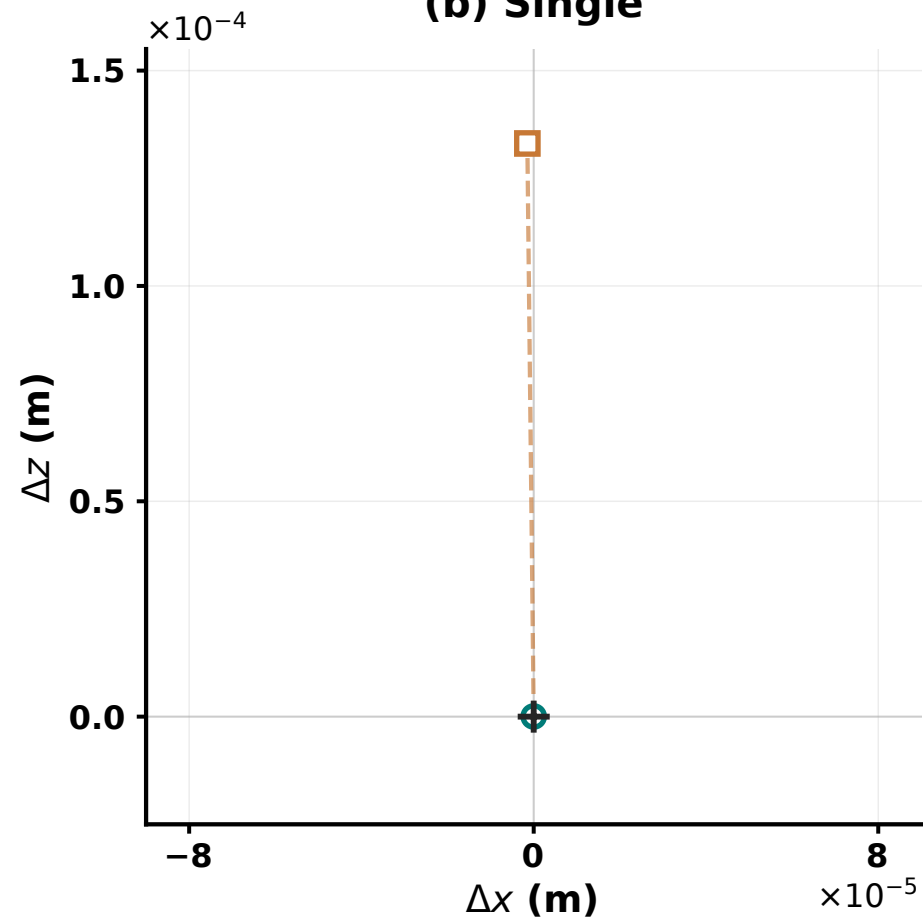
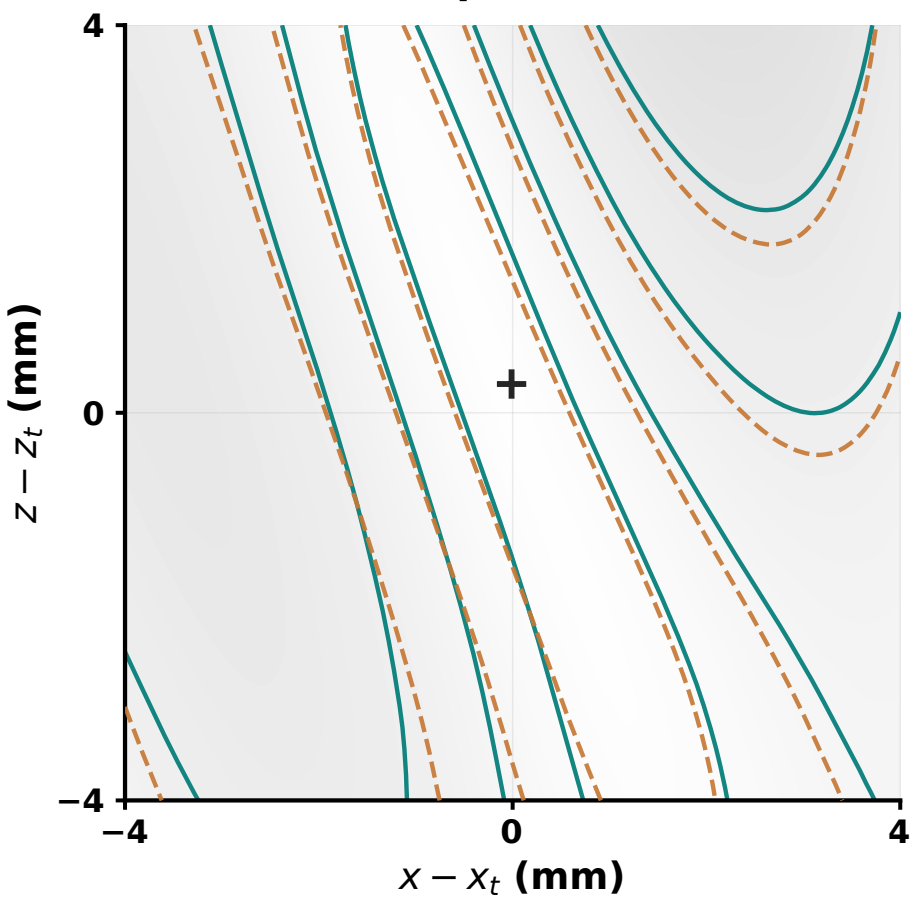
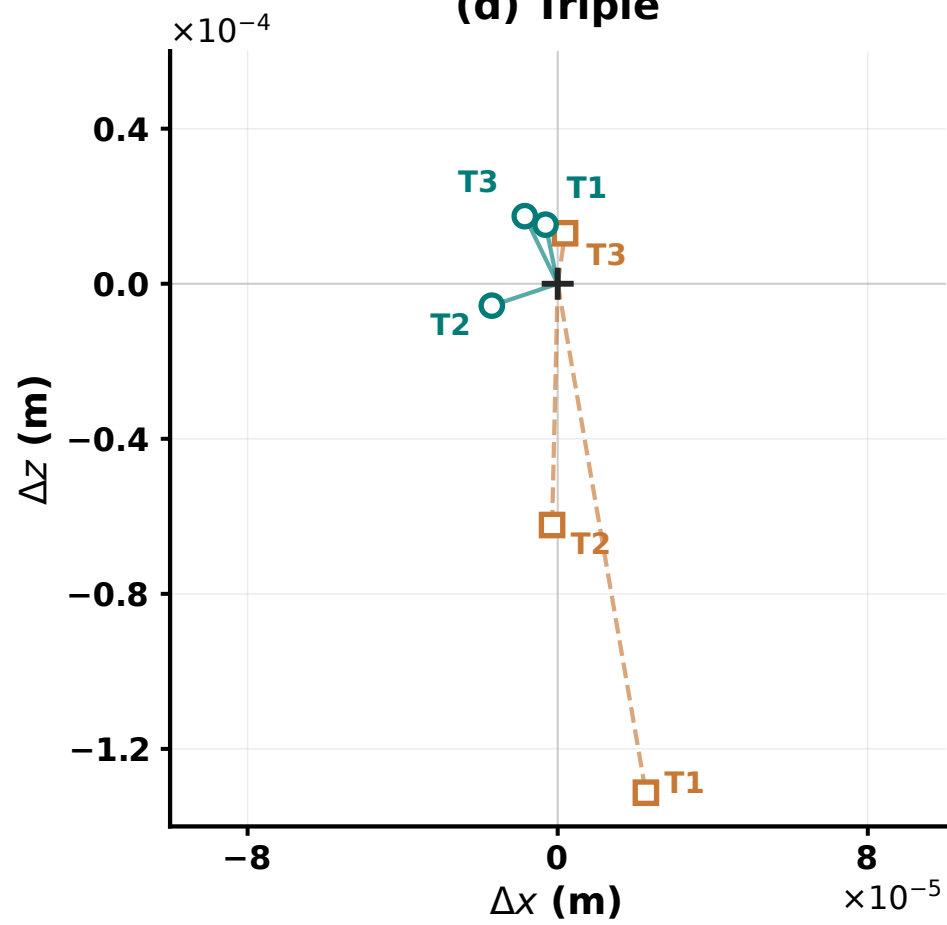




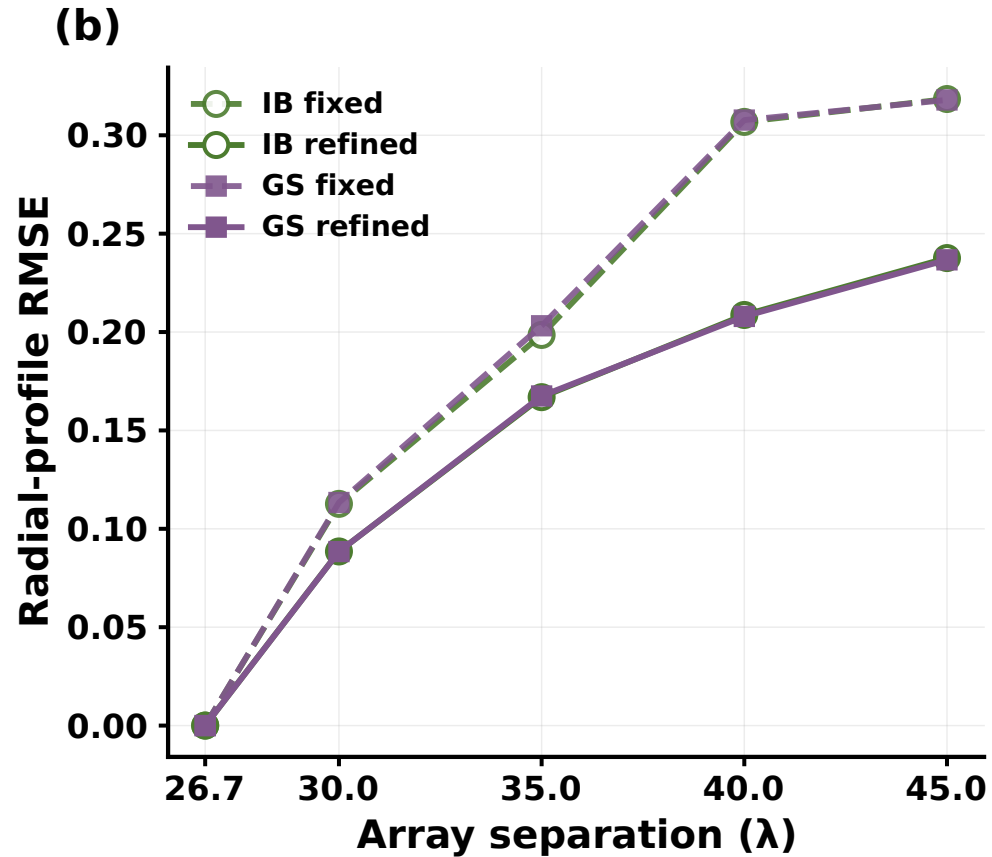
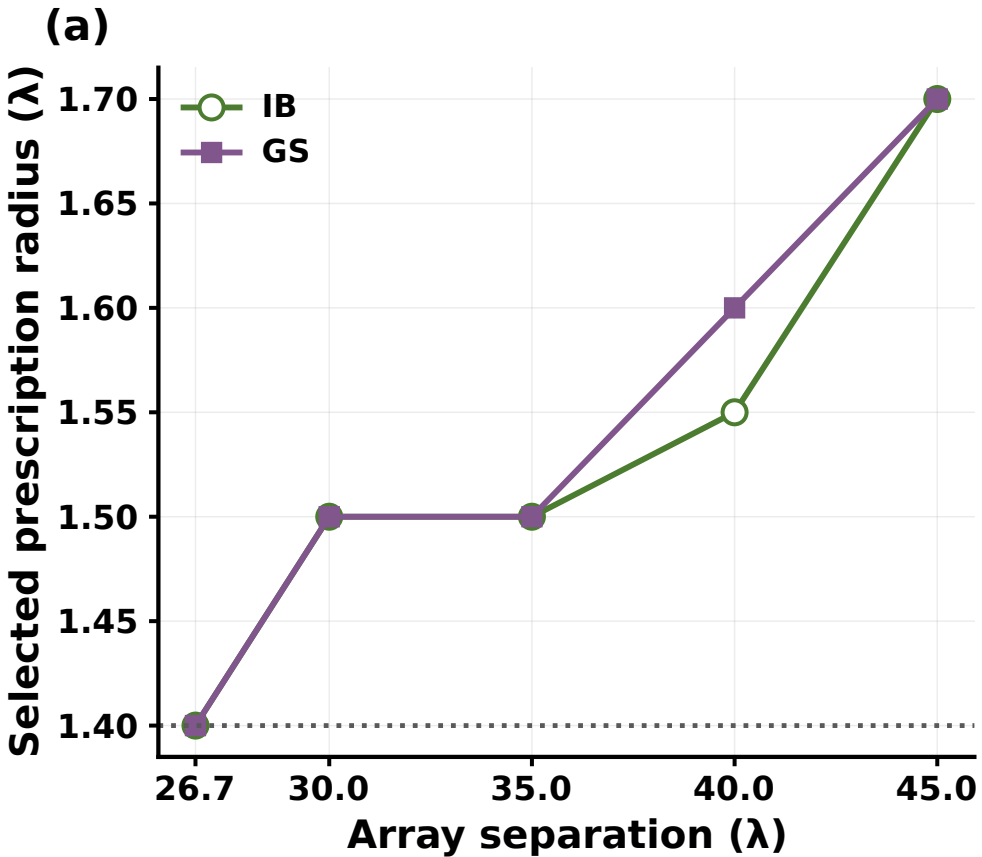


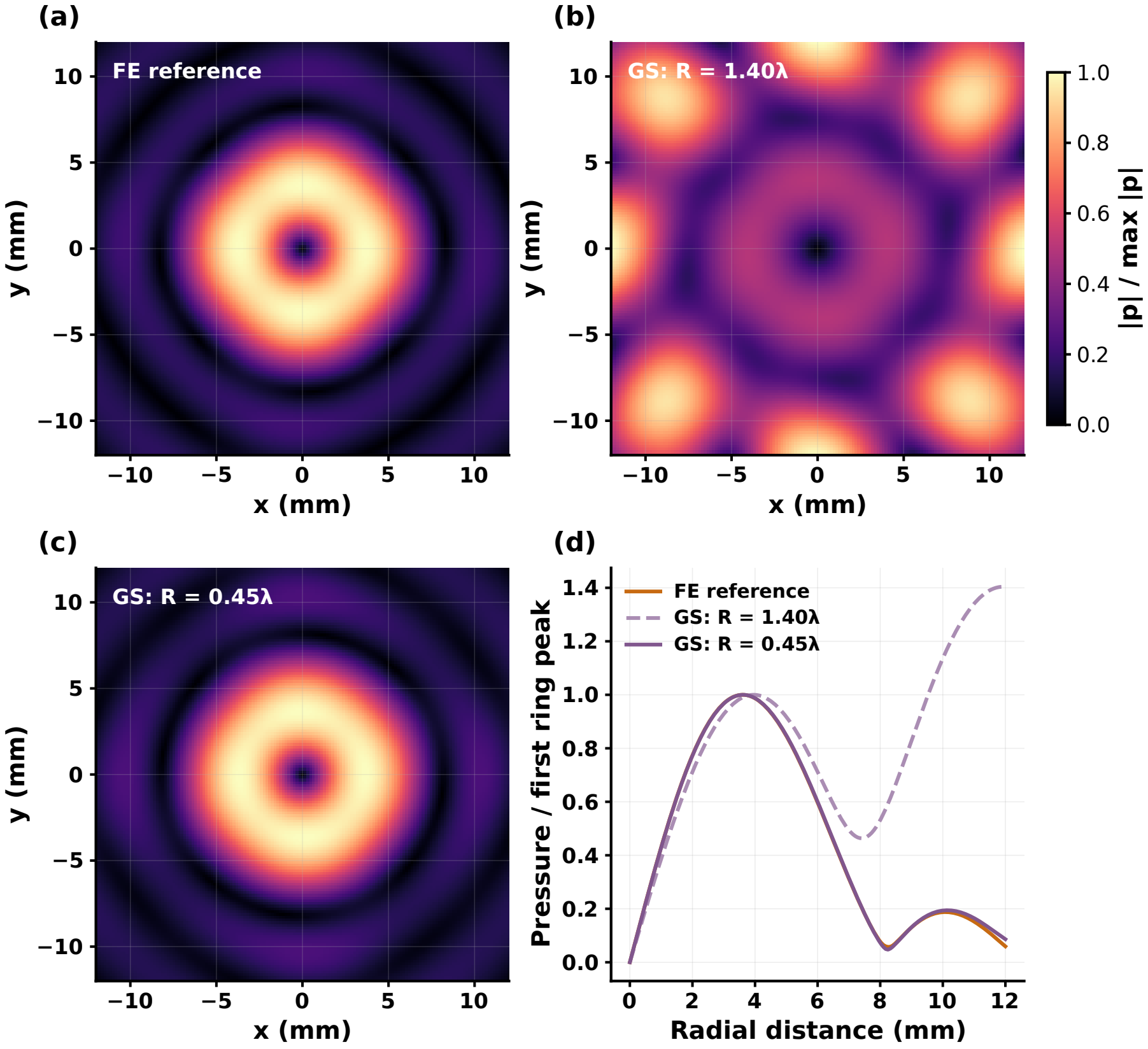


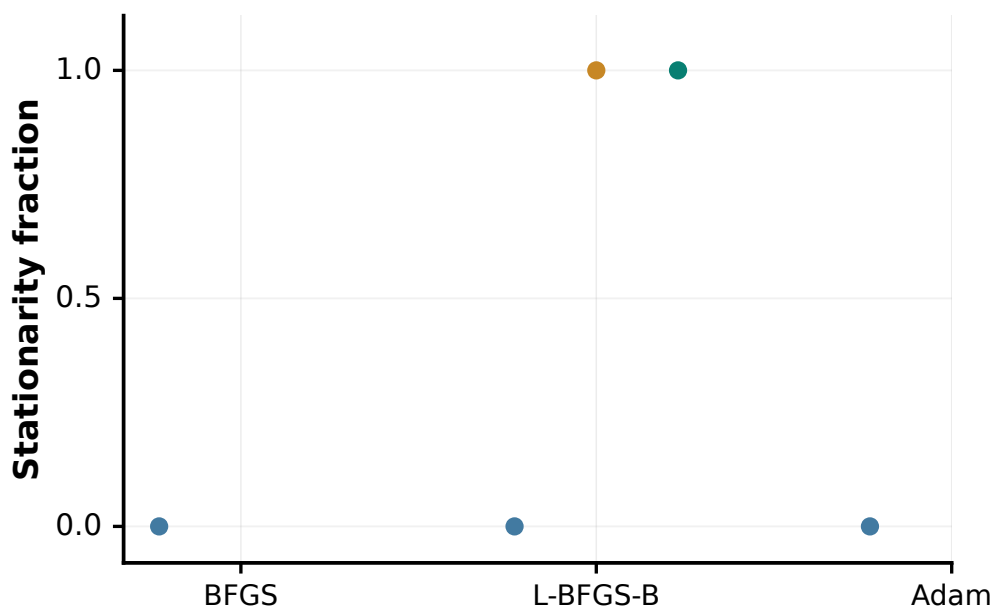
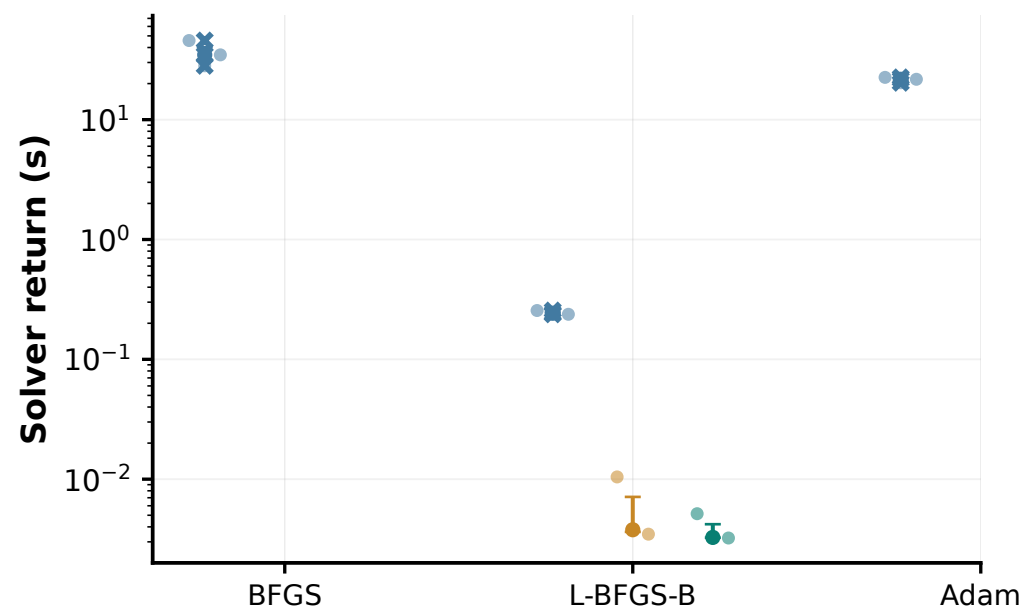
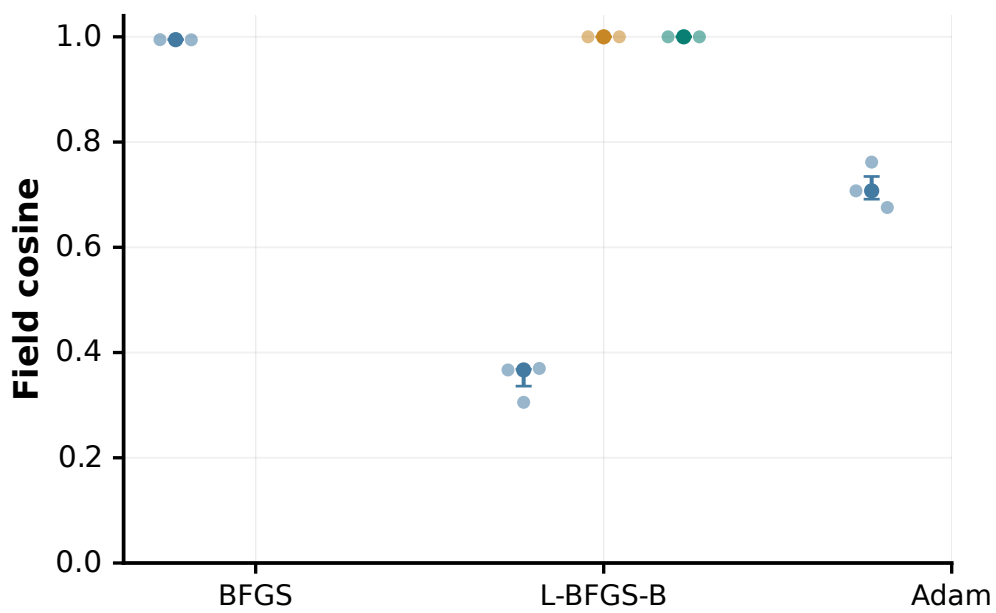
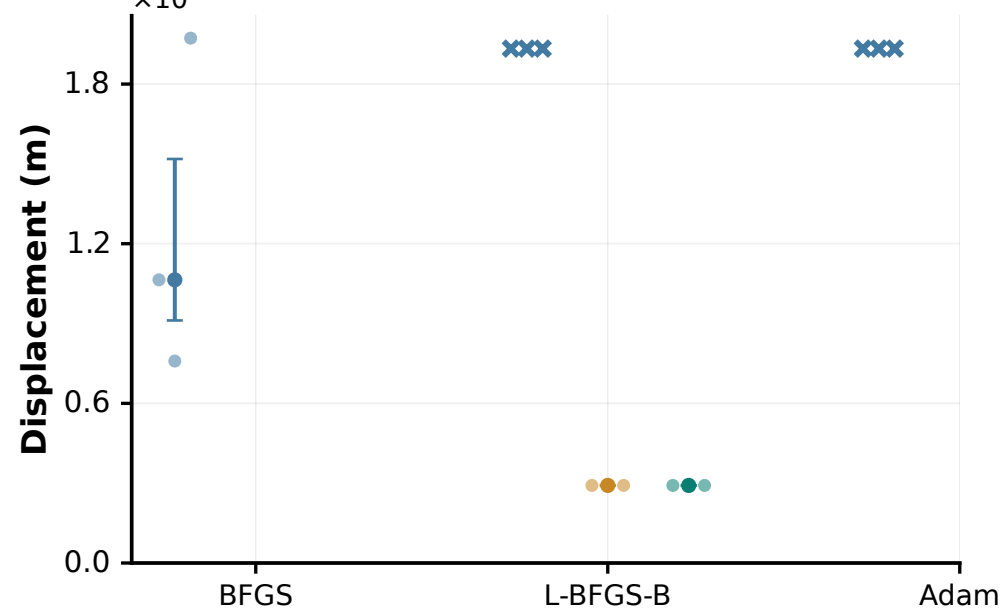


(a) Single, XZ**(b) Single****(c) Triple, T1, XZ****(d) Triple**

-□- Nearest
 -○- Mechanics offset
 + Continuous root



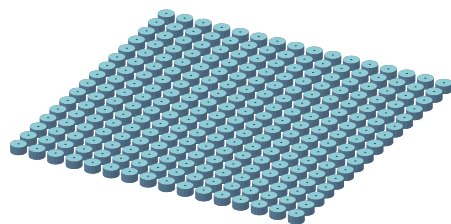


(a)**(b)****(c)****(d)**

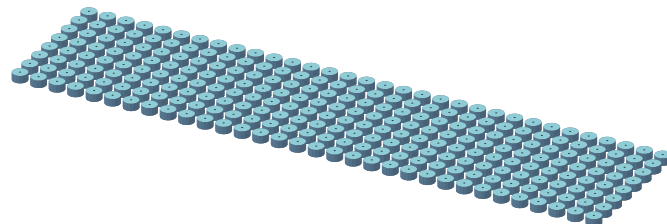
● Conventional ● FE ● GFE

× Stop unmet (b); unresolved root (d)

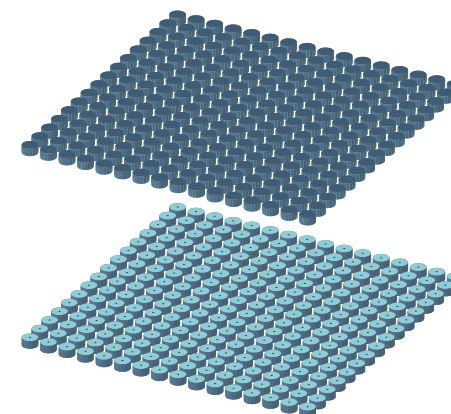
(a) Square $N = 256$



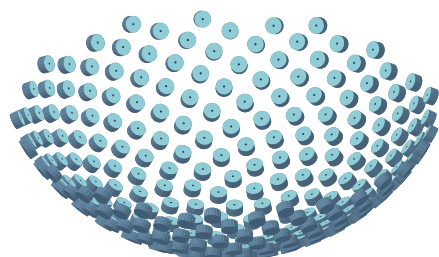
(b) Rectangular $N = 256$



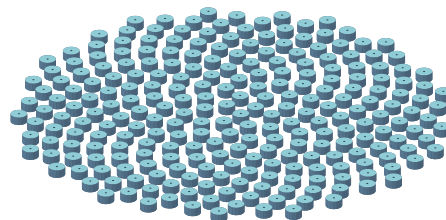
(c) Opposed $N = 512$



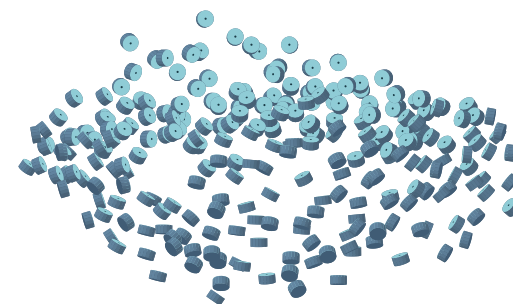
(d) Spherical cap $N = 256$



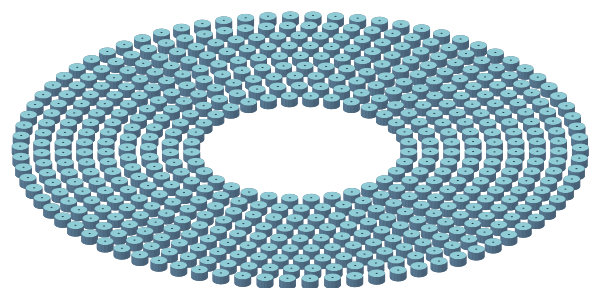
(e) Fermat disk $N = 256$



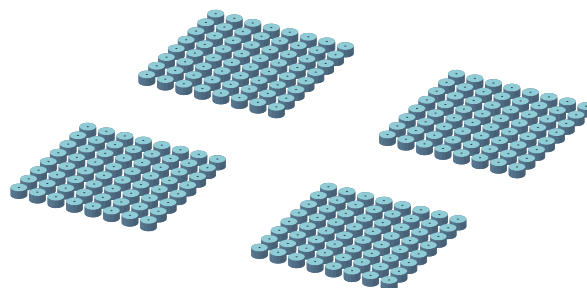
(f) Arbitrary 3D $N = 256$



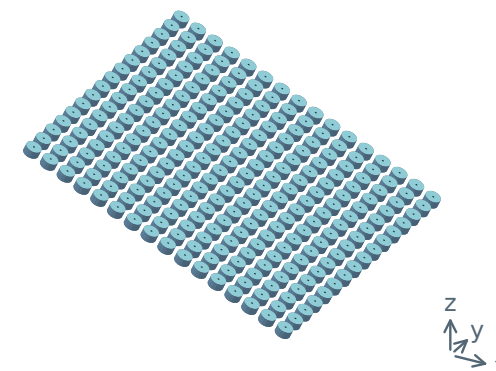
(g) Annular $N = 512$



(h) Four clusters $N = 256$



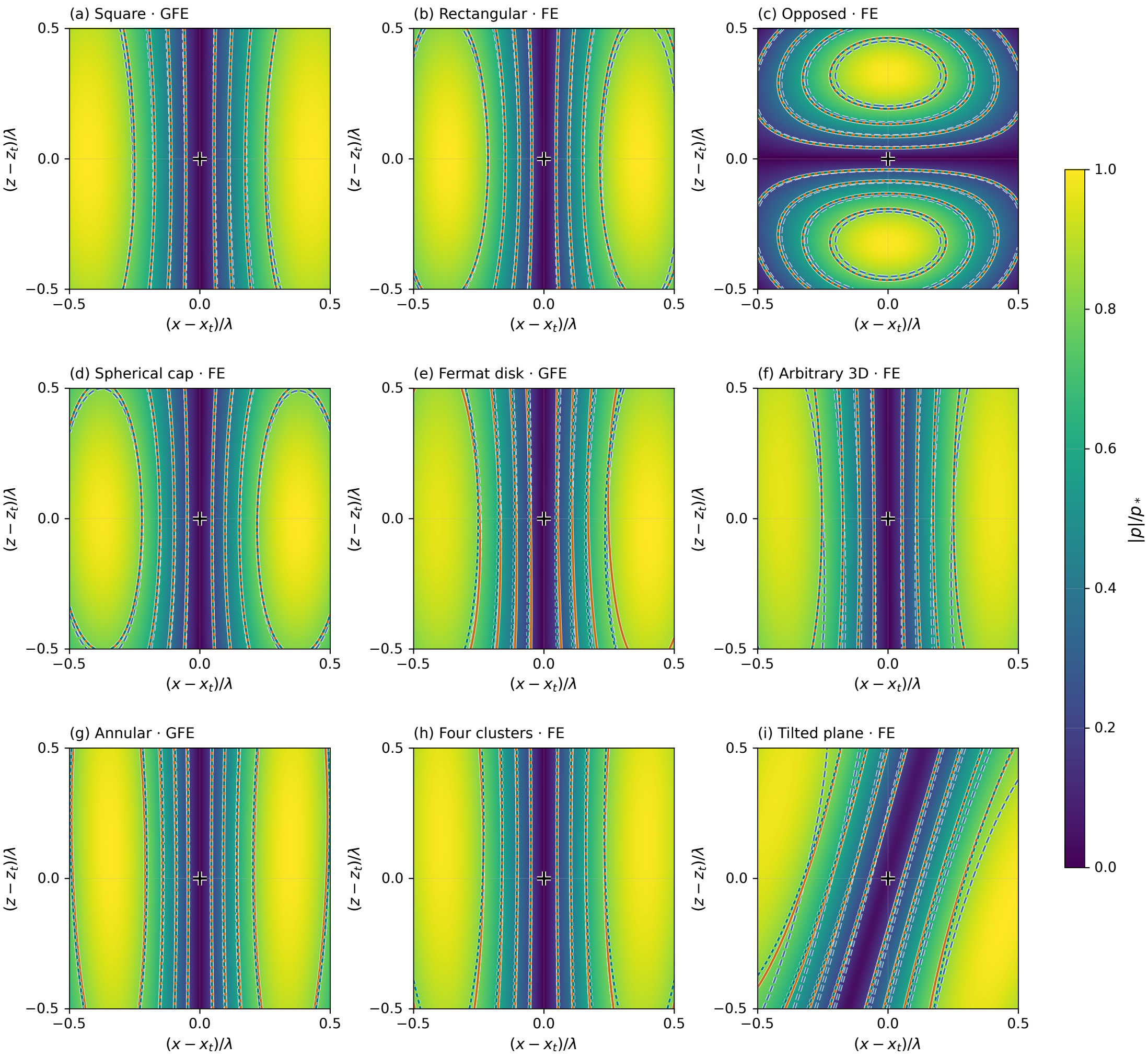
(i) Tilted plane $N = 256$



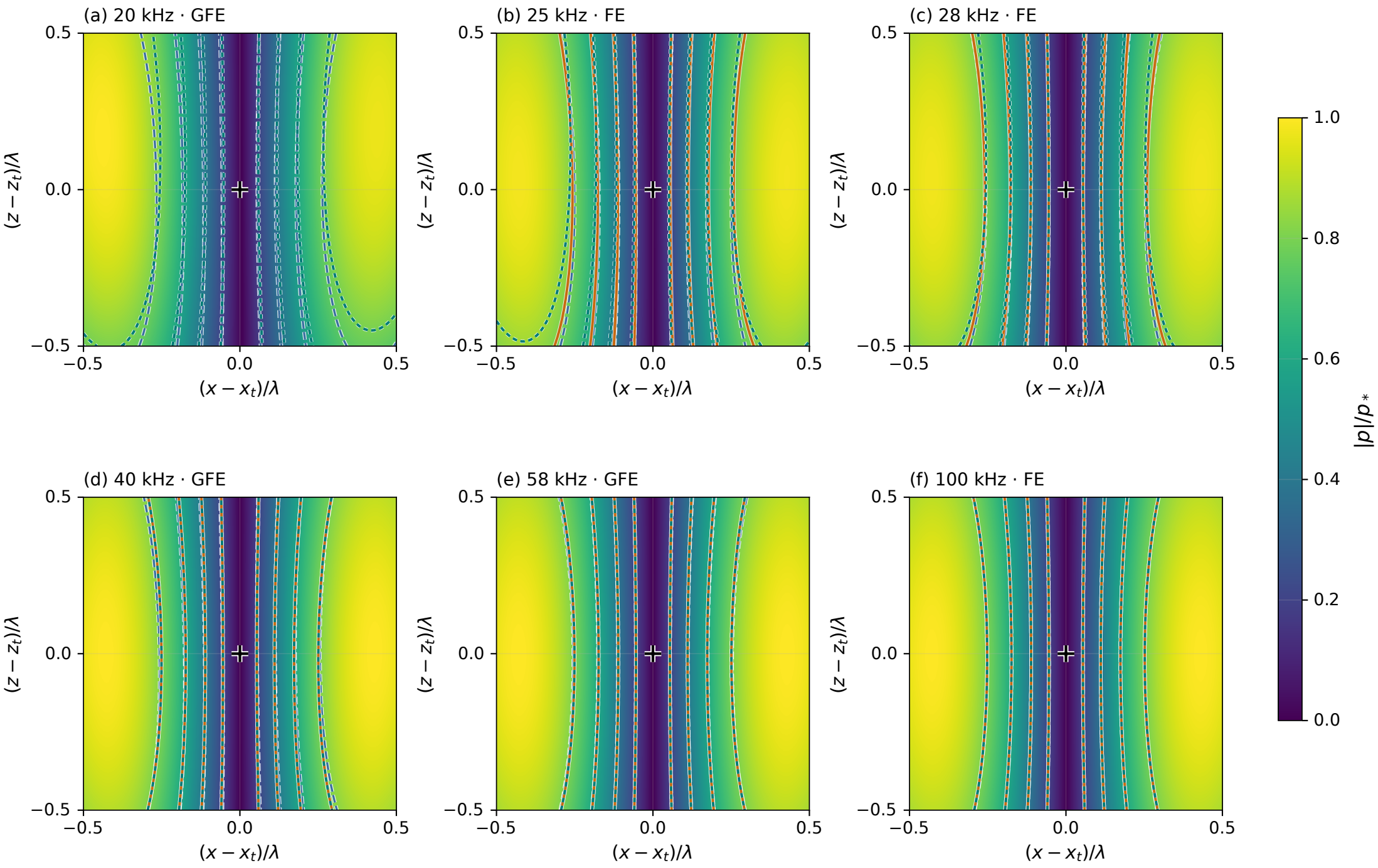
50 mm

● Emitting face ● Source point

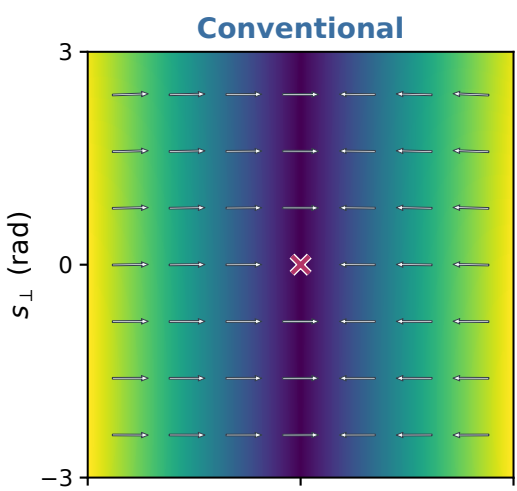
--- Conventional — FE - - - GFE



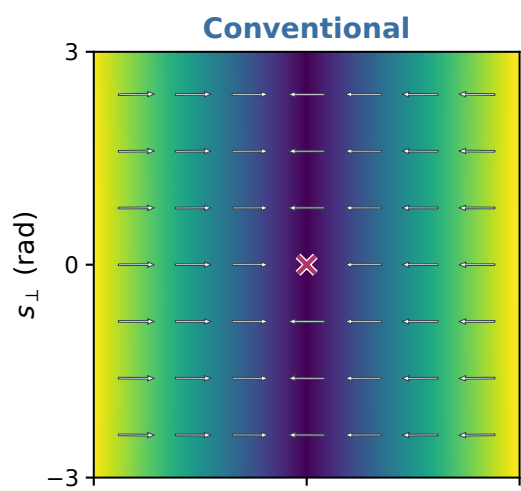
--- Conventional — FE - - - GFE



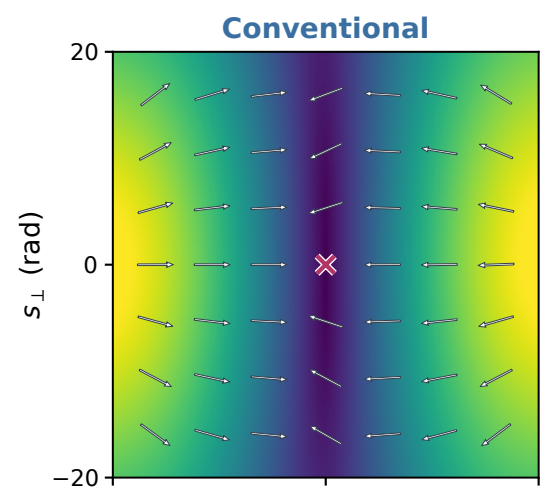
(a) Square · GFE



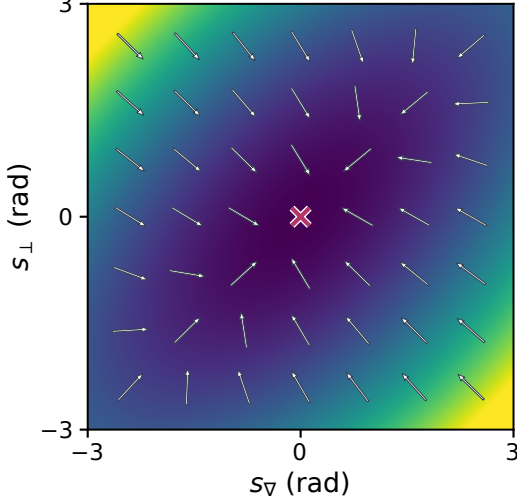
(b) Rectangular · FE



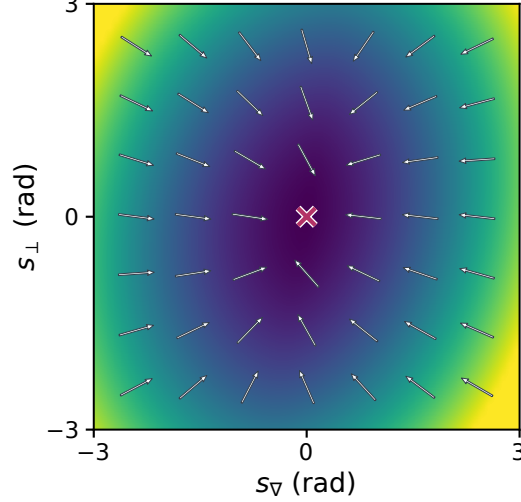
(c) Opposed · FE



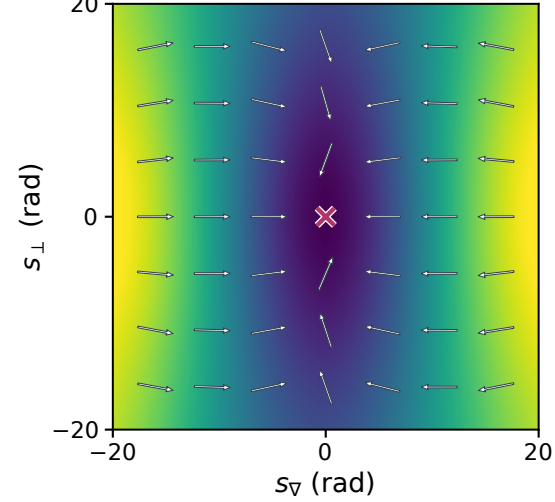
(a) Square · GFE



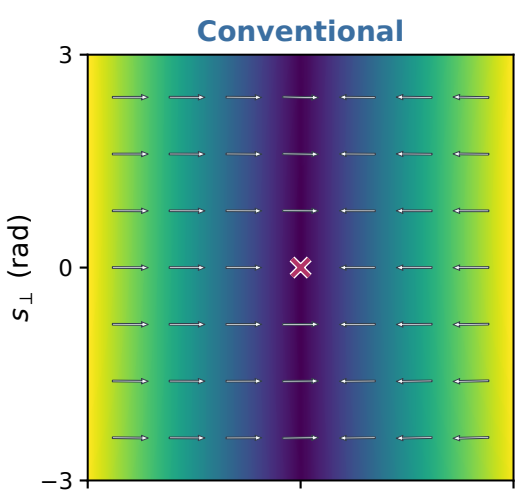
(b) Rectangular · FE



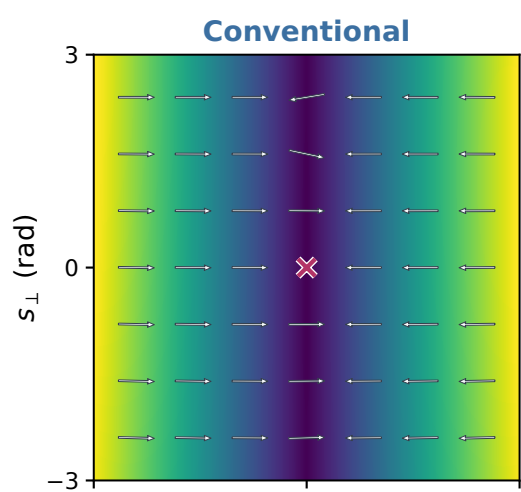
(c) Opposed · FE



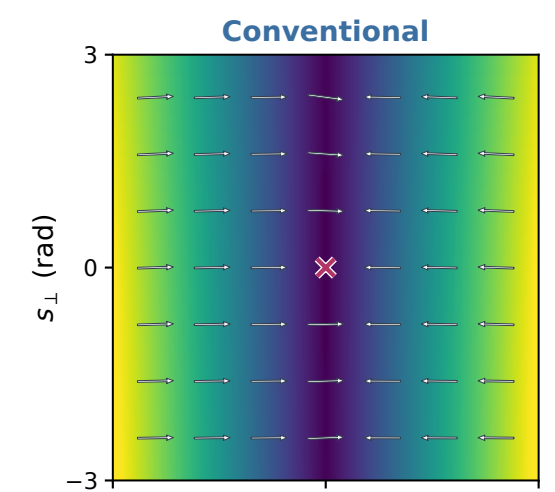
(d) Spherical cap · FE



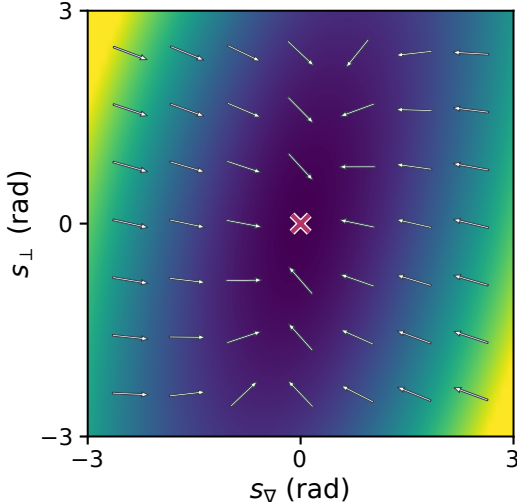
(e) Fermat disk · GFE



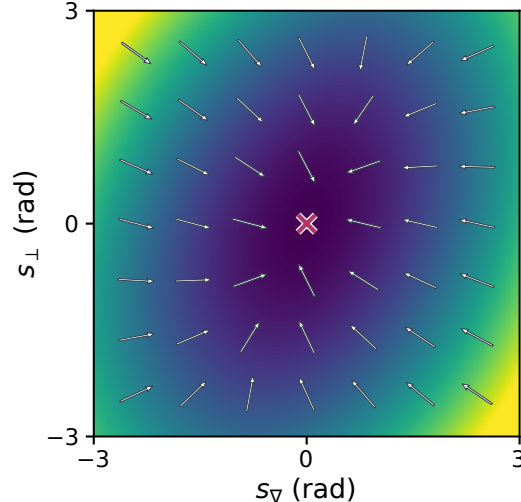
(f) Arbitrary 3D · FE



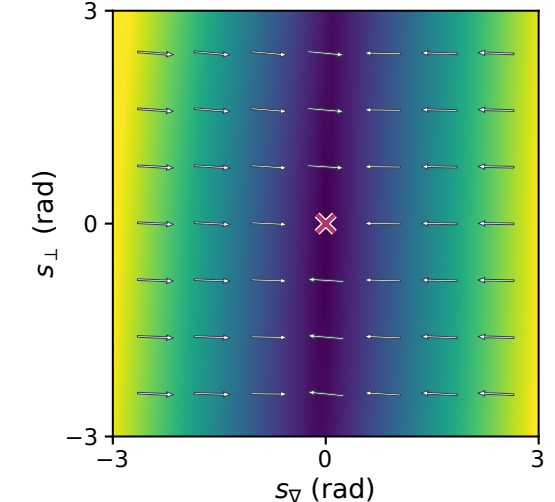
(d) Spherical cap · FE



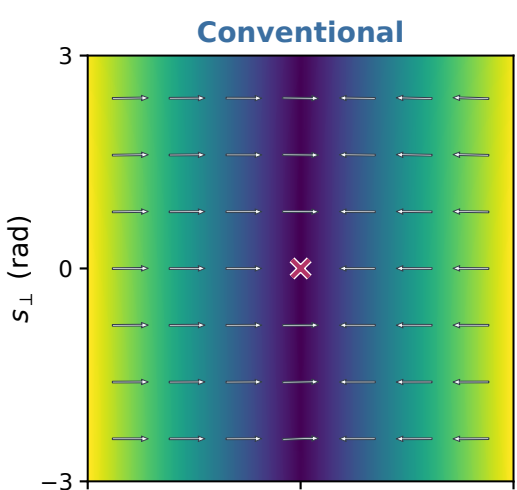
(e) Fermat disk · GFE



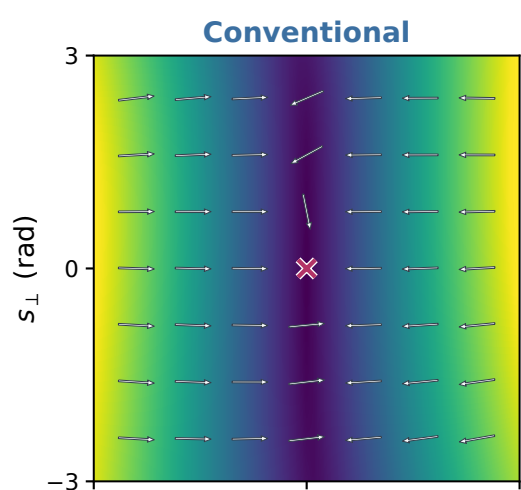
(f) Arbitrary 3D · FE



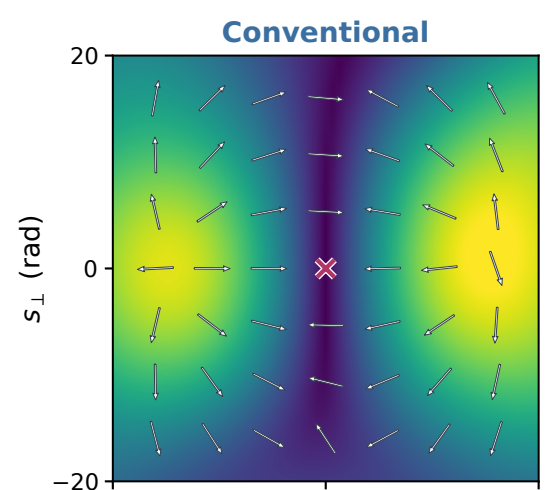
(g) Annular · GFE



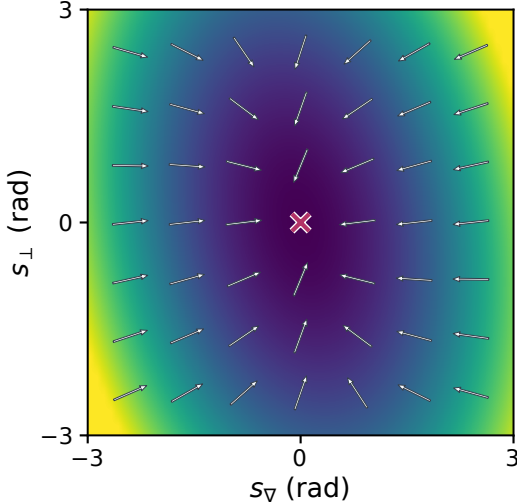
(h) Four clusters · FE



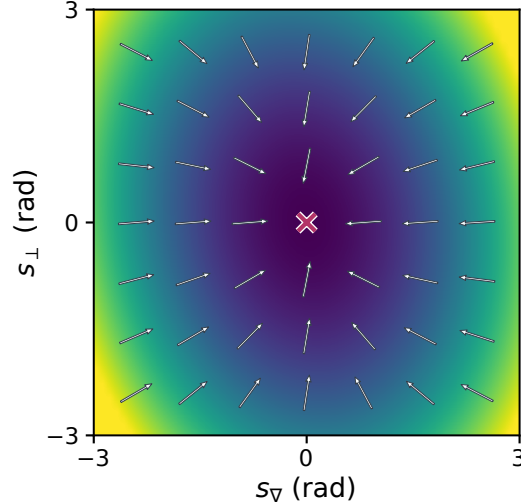
(i) Tilted plane · FE



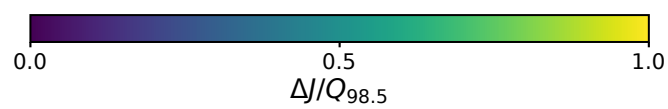
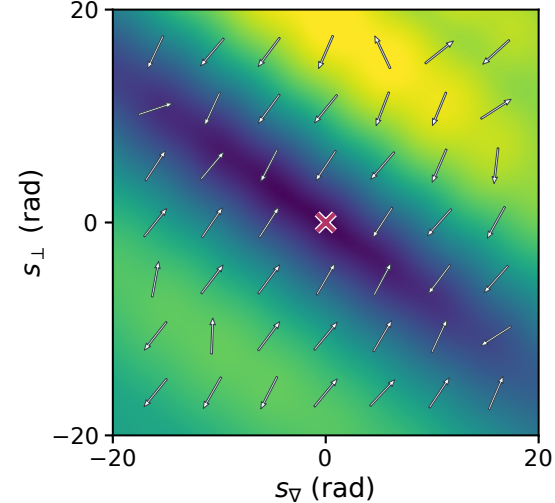
(g) Annular · GFE

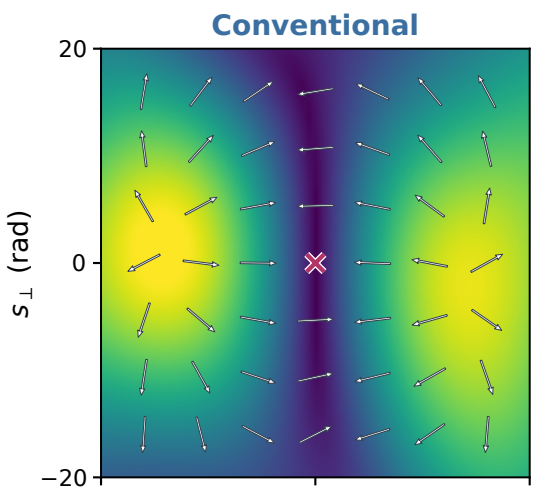
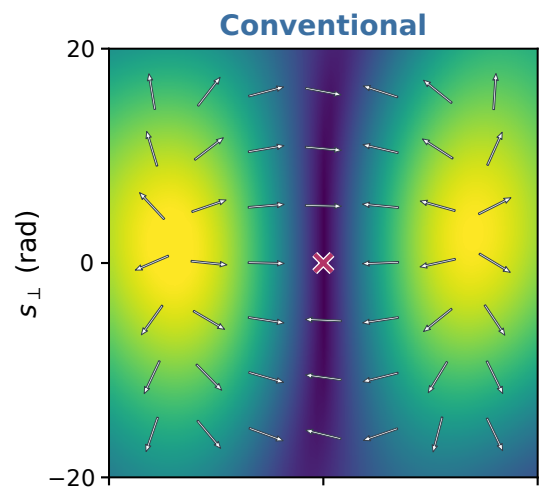
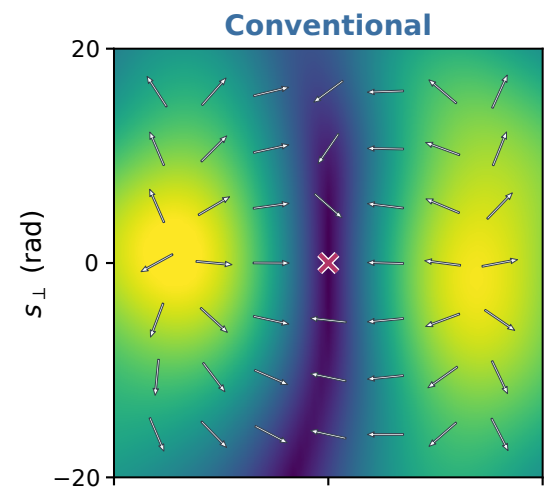
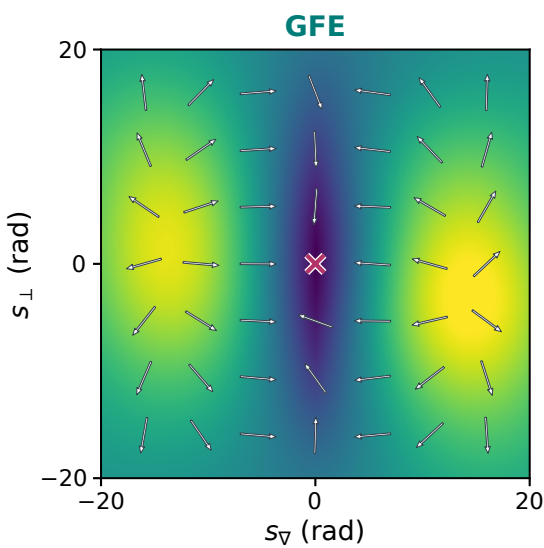
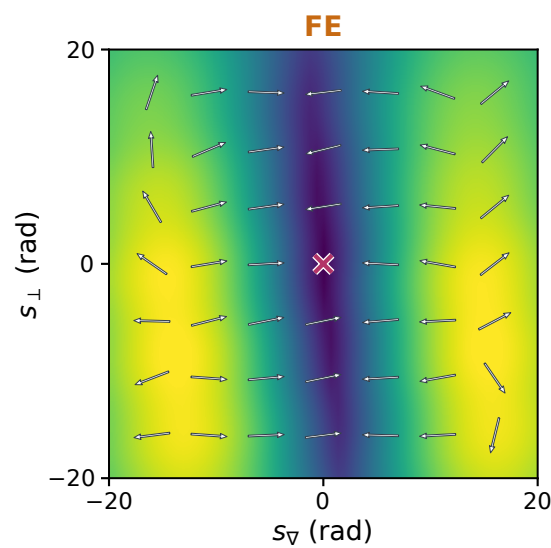
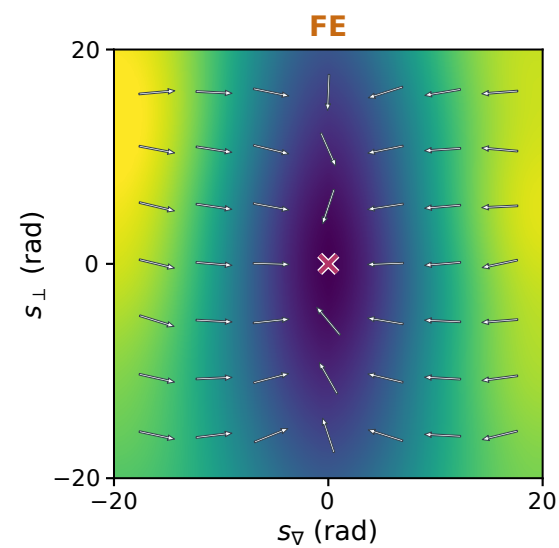
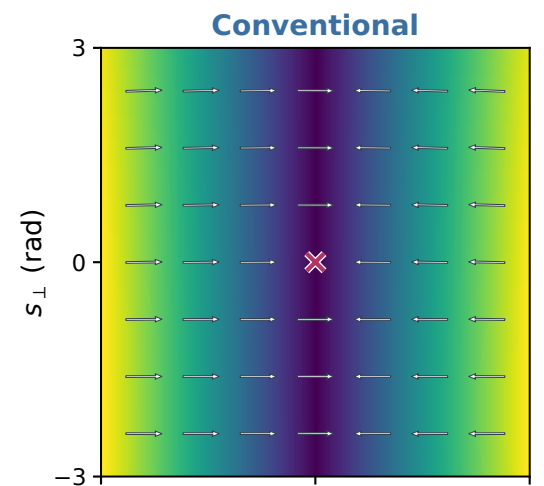
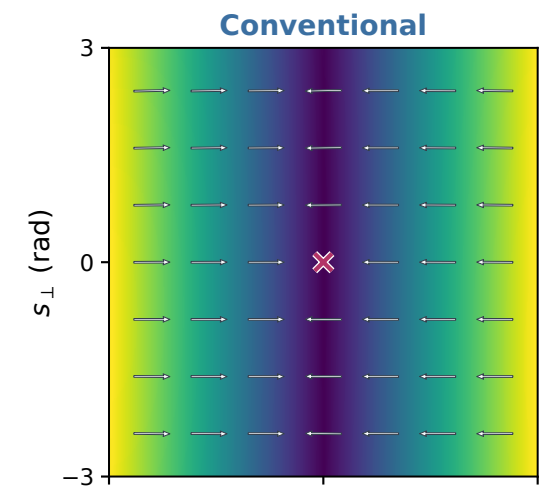
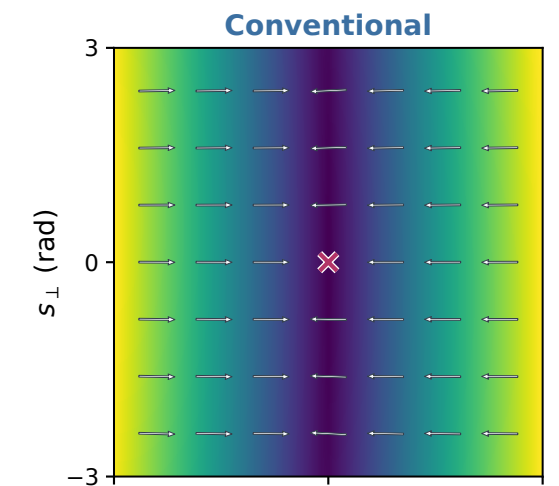
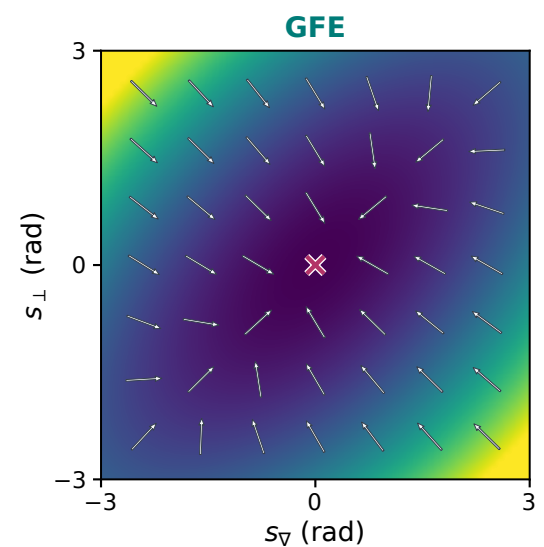
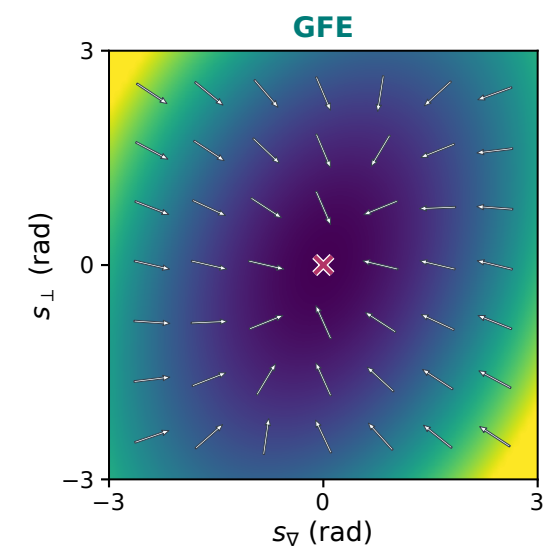
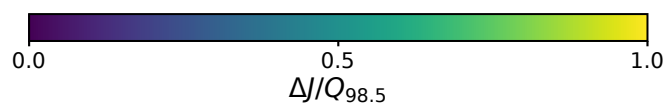
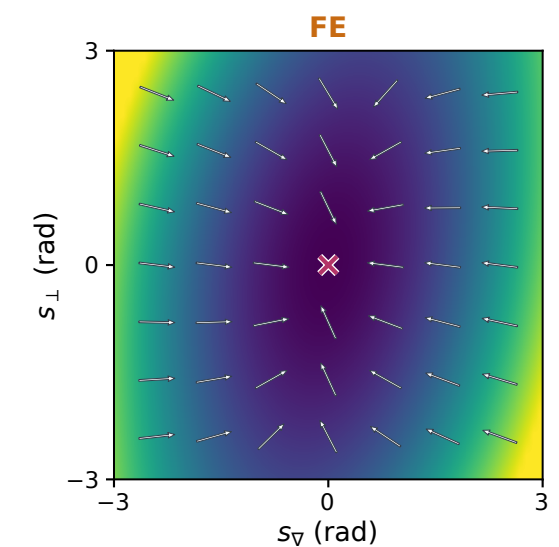


(h) Four clusters · FE

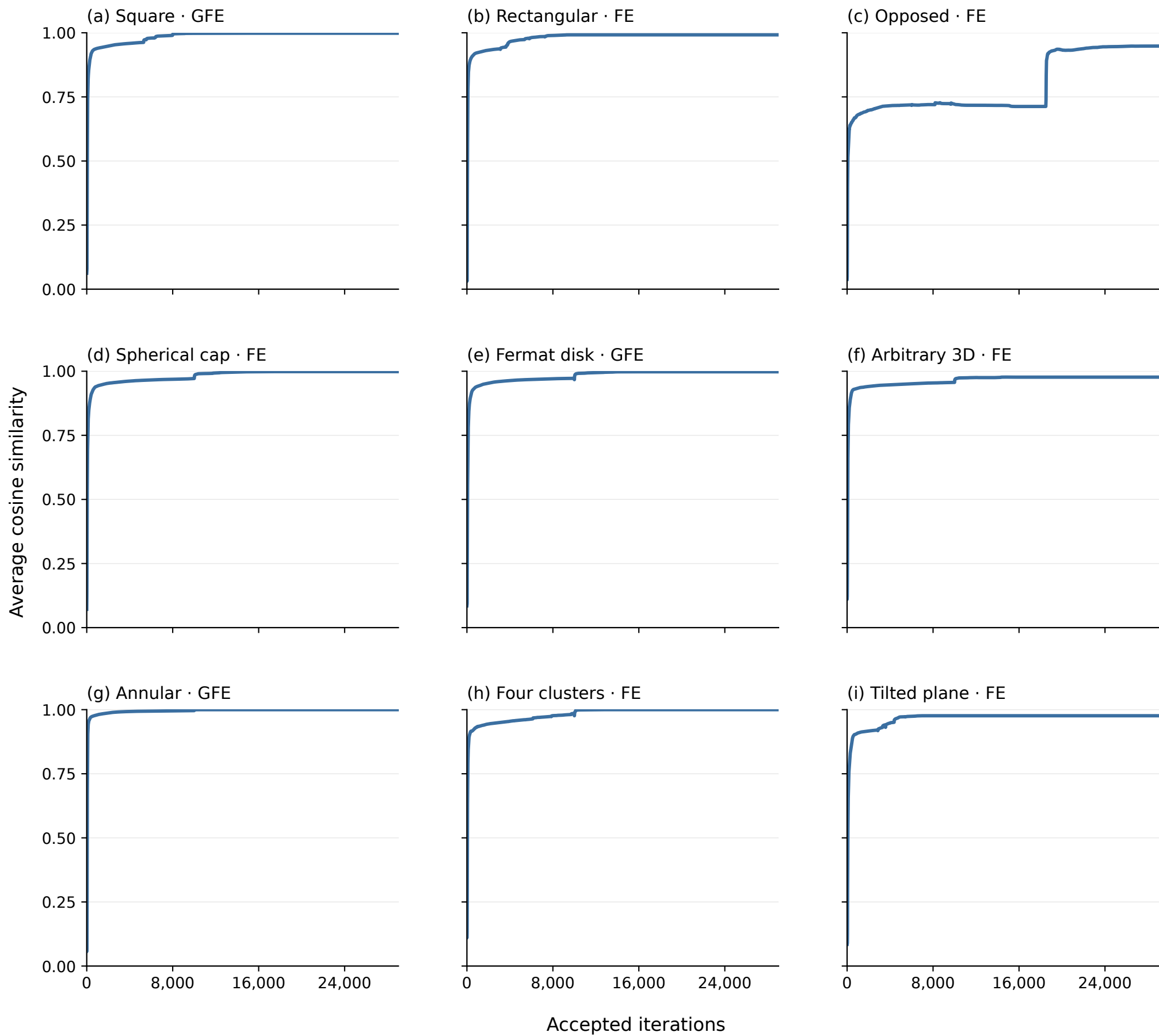


(i) Tilted plane · FE

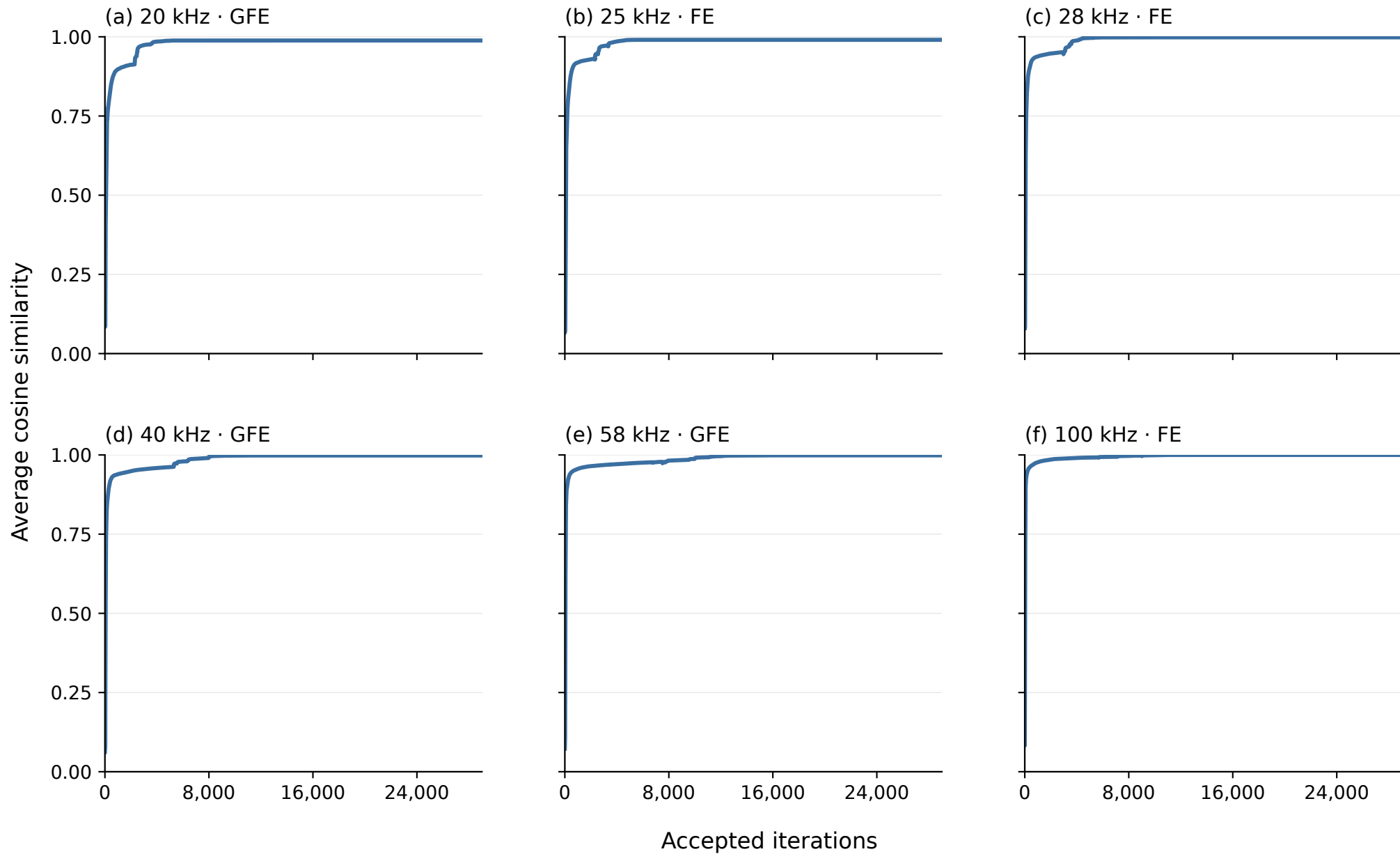


(a) 20 kHz · GFE**(b) 25 kHz · FE****(c) 28 kHz · FE****(d) 40 kHz · GFE****(e) 58 kHz · GFE****(f) 100 kHz · FE****(g) 20 kHz · GFE****(h) 25 kHz · GFE****(i) 28 kHz · GFE****(j) 40 kHz · GFE****(k) 58 kHz · GFE****(l) 100 kHz · GFE**

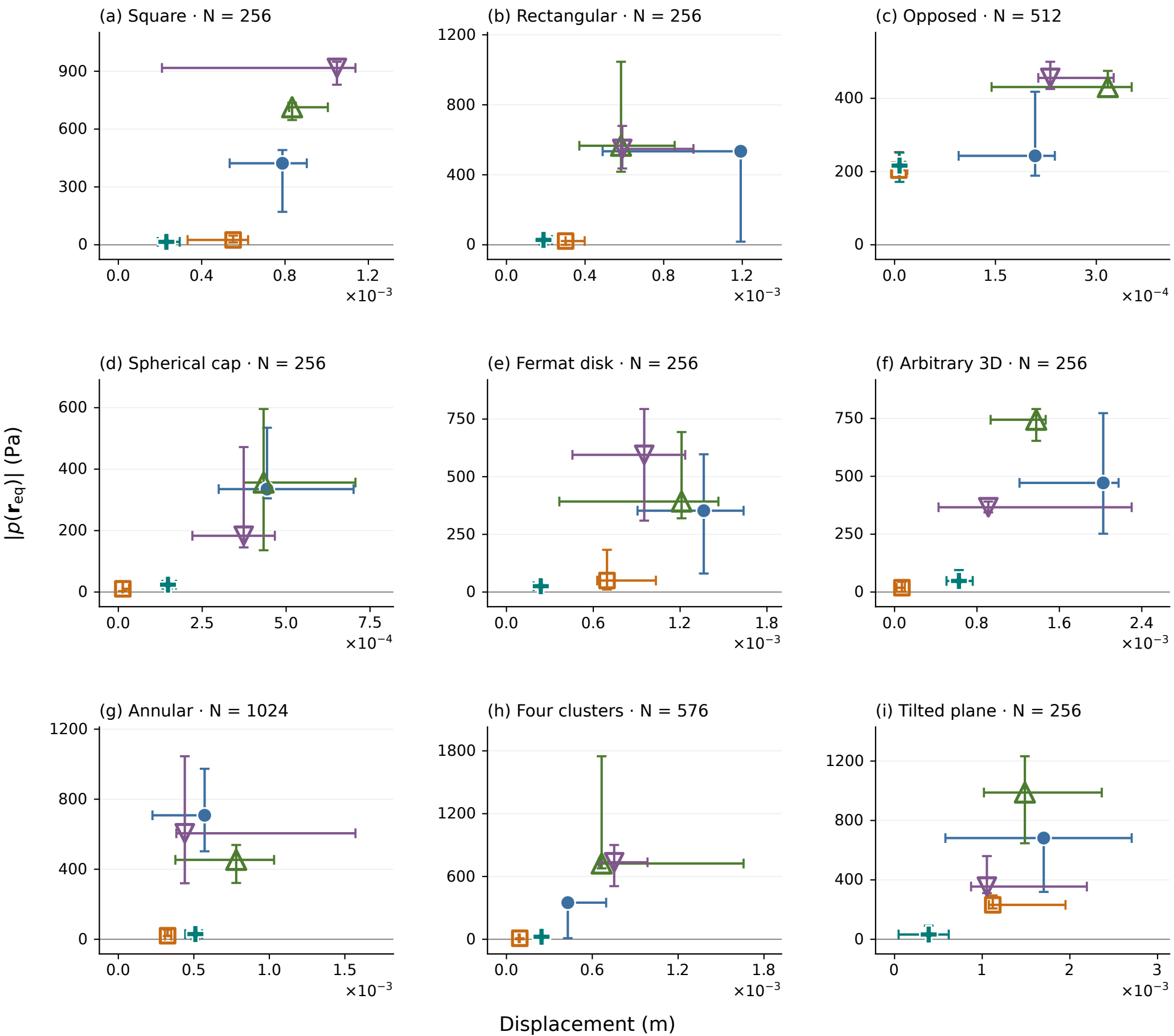
— Conventional



— Conventional

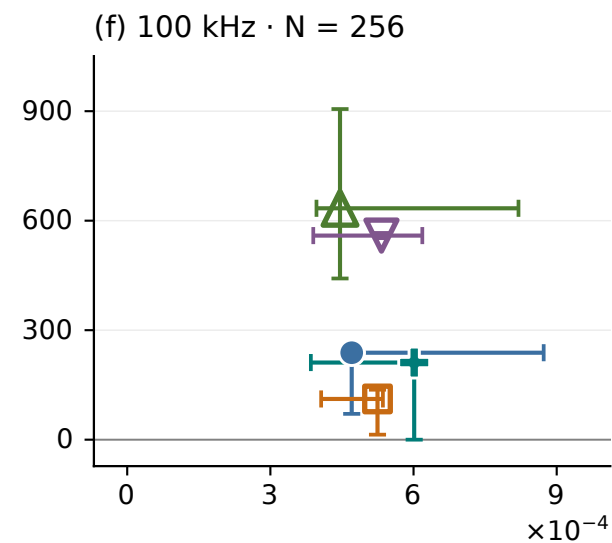
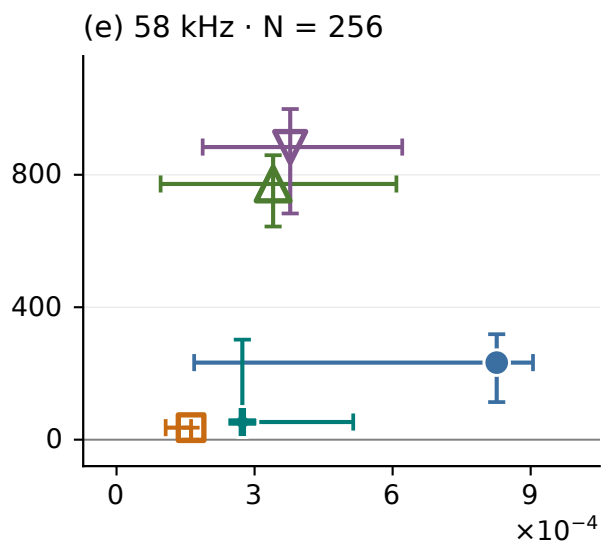
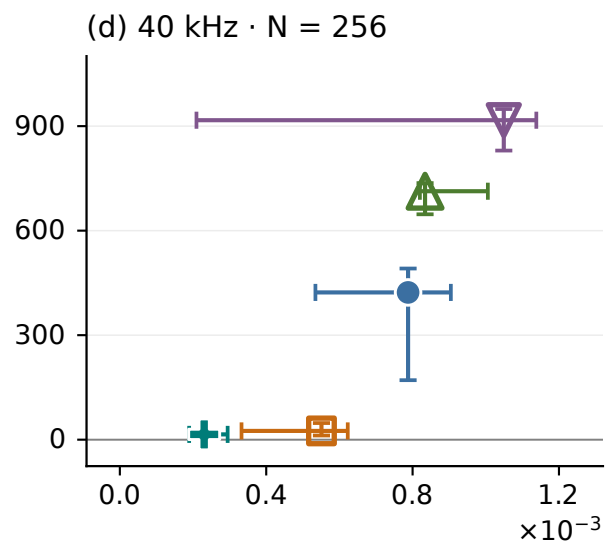
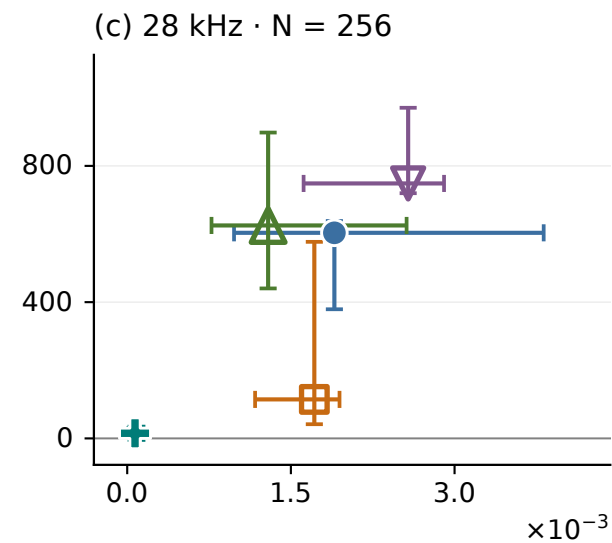
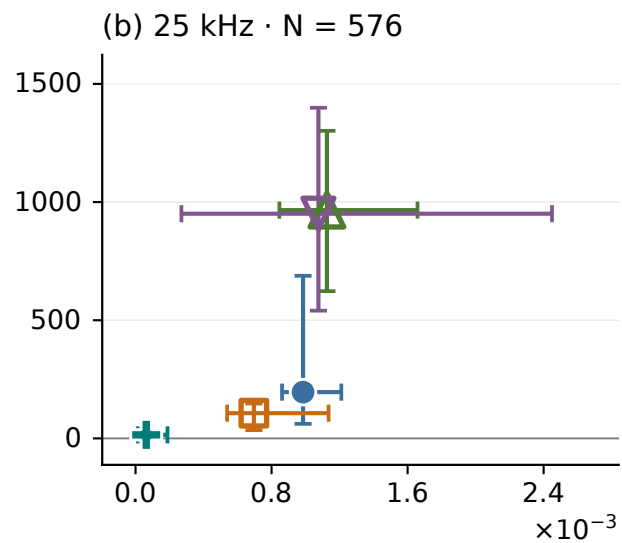
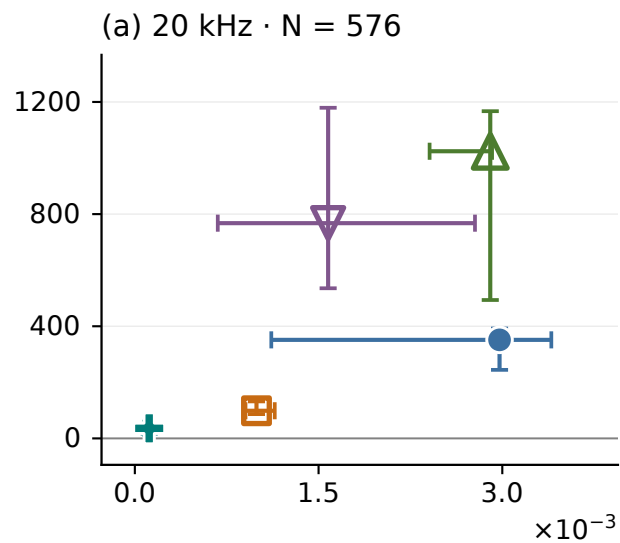


● Conventional □ FE + GFE △ IB ▽ GS



● Conventional
 □ FE
 + GFE
 △ IB
 ▽ GS

$|\rho(\mathbf{r}_{eq})|$ (Pa)



Displacement (m)